

Development, simulation & implementation of precoding techniques for satellite links

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Master's dissertation submitted in order to obtain the degree of Master of Science in Computer Science Engineering

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Ghent, May 2026

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by

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Ghent University

Faculty of Engineering and Architecture

Department of Telecommunications and Information Processing

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Keywords — Laboris magna, Laboris magna, Laboris magna, Laboris magna

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- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

An excellent style manual for science writers is [1].

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Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is “Heading 5”. Use “figure caption” for your Figure captions, and “table head” for your table title. Run-in heads, such as “Abstract”, will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

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TABLE I
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^aSample of a Table footnote.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present



Fig. 1. Example of a figure caption.

them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

ACKNOWLEDGMENT

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- [1] A. I. Perez-Neira, M. A. Vazquez, M. B. Shankar, S. Maleki, and S. Chatzinotas, “Signal processing for high-throughput satellites: Challenges in new interference-limited scenarios,” *IEEE Signal Processing Magazine*, vol. 36, no. 4, pp. 112–131, 2019.

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Symbols and Notations

Abbreviations

acf	autocorrelation function
AWGN	additive white Gaussian noise
BS	base station
BER	bit error rate
CS	circularly symmetric
CSI	channel state information
DFT	discrete Fourier transform
EVD	eigenvalue decomposition
FFT	fast Fourier transform
FIR	finite impulse response
IBR	information bit rate
i.i.d.	independent and identically distributed
KKT	Karush-Kuhn-Tucker
MI	mutual information
MIMO	multiple-input multiple-output
MMSE	minimum mean squared error
MSE	mean squared error
MU	multi-user
pdf	probability density function
PAM	pulse amplitude modulation
PSD	power spectral density
PSK	phase shift keying

QAM	quadrature amplitude modulation
QoS	Quality-of-Service
RX	receiver
SISO	single-input single-output
SNR	signal-to-noise ratio
SR	sum rate
SU	single-user
SVD	singular value decomposition
TX	transmitter
UT	user terminal
WMSE	weighted mean squared error
WMMSE	weighted minimum mean squared error
WSR	weighted sum rate

Mathematical Notations

TO DO

System Parameters

N_T	Number of transmit antennas at the BS
N_R	Number of receive antennas at each UT
N_S	Number of data streams per UT
K	Number of UTs

Symbols

TO DO

Chapter 1

Introduction

Part I

Background Material

Chapter 2

Communication Systems

This chapter establishes the system models and information-theoretic framework that underlie the analysis presented throughout this work. We begin by describing the general architecture of a digital communication system and then specialize to the system model adopted in this work. Finally, we review the fundamental information-theoretic concepts required to analyze the performance of these systems.

2.1 Digital Communication Systems

A general digital communication system transfers digital information from a source to a destination over a physical transmission medium, referred to as the channel. The corresponding block diagram is shown in Fig. 2.1.

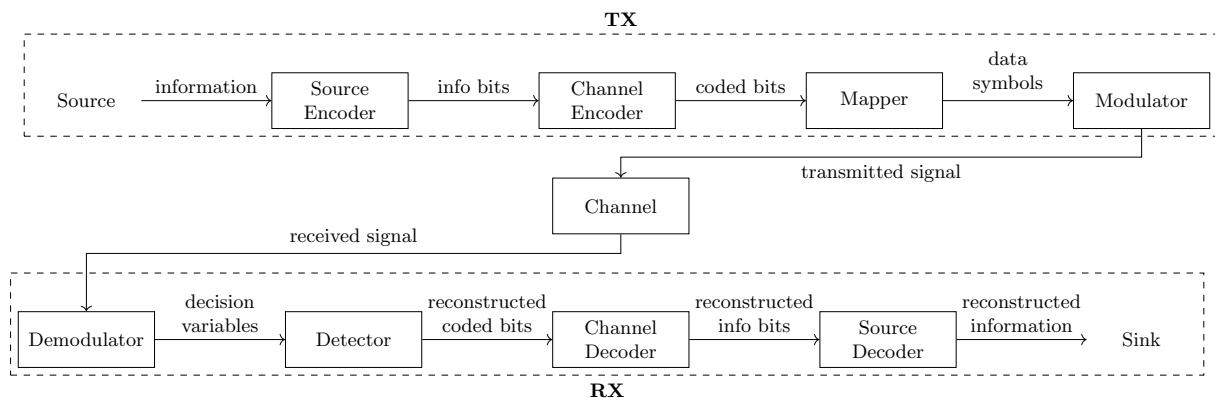


Figure 2.1: Block diagram of a general digital communication system.

At the transmitter (TX), the information sequence is first processed by a source encoder to remove redundancy, followed by a channel encoder that introduces controlled redundancy to enable error detection. The resulting coded bit sequence is partitioned into blocks of m_c bits, each of which is mapped onto a complex-valued symbol drawn from a constellation $\mathcal{X}$ of size 2^{m_c} . The modulator then converts the symbol sequence into a baseband signal suitable for transmission.

During transmission, the channel distorts the signal due to attenuation, multipath propagation, and additive noise, such that the received signal differs from the transmitted signal.

At the receiver (RX), the demodulator converts the received signal into a sequence of decision variables. These are processed by the detector to obtain estimates of the originally transmitted symbols, which are then mapped back to coded bits. Finally, the channel and source decoders reconstruct the original information sequence.

In this work, we consider a simplified version of this general architecture in order to focus on the aspects most relevant to our study. Source coding and decoding are omitted, and the information is assumed to be available as a sequence of independent and identically distributed (i.i.d.) bits without prior compression or digitization. Channel coding and decoding are also not considered, such that the analysis is restricted to uncoded transmission.

The transmission medium is assumed to be a wireless channel affected by fading and additive white Gaussian noise. Furthermore, the channel is considered to be frequency-flat, meaning that its response remains constant over the entire signal bandwidth. The specific channel models used in this work are introduced in the relevant chapters.

2.2 MIMO Communication Systems

When both the TX and the RX are equipped with multiple antennas, the system is referred to as a multiple-input multiple-output (MIMO) system.

Compared to a single-antenna link, the multiple antennas provide additional spatial degrees of freedom that can be exploited to increase data throughput, improve reliability, or achieve a trade-off between both. Spatial multiplexing techniques enable significant capacity gains by transmitting multiple independent data streams simultaneously over different antennas. A well-known example is the BLAST architecture [1]. Alternatively, spatial diversity can be achieved through space-time coding schemes, which distribute symbols across both space and time to improve robustness against fading. A classical example is the Alamouti scheme [2]. Throughout this work, the focus is on spatial multiplexing and, in particular, on how the simultaneous transmission of multiple data streams over a shared channel can be made most efficient through precoding.

2.2.1 SU-MIMO System Model

In the point-to-point case, where a single TX with N_T antennas communicates with a single RX with N_R antennas, the system is called a single-user (SU) MIMO system. A block diagram is shown in Fig. 2.2.

The discrete-time complex baseband input-output relation of a SU-MIMO Gaussian channel is characterized by

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad (2.1)$$

where $\mathbf{x} \in \mathbb{C}^{N_T}$ is the transmitted signal vector, $\mathbf{y} \in \mathbb{C}^{N_R}$ is the received signal vector, $\mathbf{H} \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix, and $\mathbf{n} \in \mathbb{C}^{N_R}$ is the additive noise vector. The (n_r, n_t) -th entry of $\mathbf{H}$ represents the complex channel coefficient between transmit antenna n_t and receive

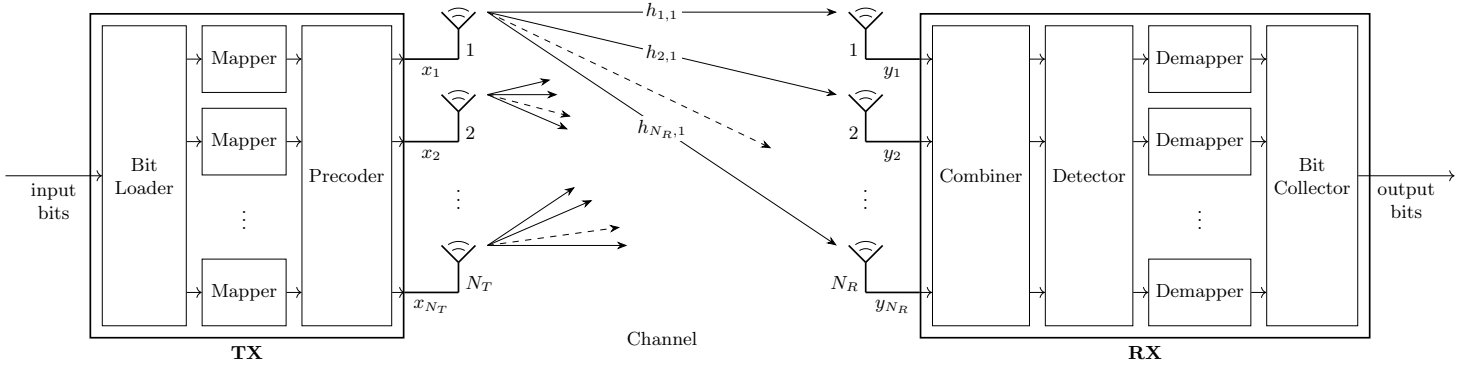


Figure 2.2: Block diagram of a SU-MIMO system.

antenna n_r . The noise vector is circularly symmetric (CS) complex Gaussian distributed, i.e. $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}_{N_R})$, such that its entries are spatially independent, each with an expected power of N_0 . The time index is omitted for brevity.

At the TX, the information bits are divided into N_S streams and mapped onto data symbols, which are collected in the vector $\mathbf{a} \in \mathbb{C}^{N_S \times 1}$. Then, the transmitted signal is obtained by mapping the N_S data symbol streams onto the N_T transmit antennas according to a linear precoding scheme, represented by a matrix $\mathbf{F} \in \mathbb{C}^{N_T \times N_S}$. The transmitted signal can therefore be expressed as

$$\mathbf{x} = \mathbf{F} \mathbf{a}. \quad (2.2)$$

The data symbols are assumed to be zero-mean, mutually uncorrelated, and normalized, such that $\mathbb{E}[\mathbf{a}\mathbf{a}^H] = \mathbf{I}_{N_S}$. Throughout this work, the power allocation is assumed to be incorporated into the precoding matrix, unless explicitly stated otherwise. Under this convention, the total transmit power is constrained by (see Appendix E)

$$\mathbb{E}[\|\mathbf{x}\|^2] = \text{tr}(\mathbf{F}^H \mathbf{F}) \leq P_t. \quad (2.3)$$

At the RX, the N_R received signals are linearly combined to obtain the decision-variable vector $\mathbf{z} \in \mathbb{C}^{N_S \times 1}$. The combiner is represented by a matrix $\mathbf{G} \in \mathbb{C}^{N_S \times N_R}$. The decision variables can therefore be expressed as

$$\mathbf{z} = \mathbf{G} \mathbf{y}. \quad (2.4)$$

Based on $\mathbf{z}$, the detector forms estimates of the transmitted data symbols, which are then mapped back to bits.

Substituting (2.2) into (2.1) and applying (2.4) yields the relation between the transmitted data symbols and the decision variables:

$$\mathbf{z} = (\mathbf{G} \mathbf{H} \mathbf{F}) \mathbf{a} + \mathbf{G} \mathbf{n}, \quad (2.5a)$$

$$= \mathbf{T} \mathbf{a} + \mathbf{n}_\mathbf{G}, \quad (2.5b)$$

where $\mathbf{T} \in \mathbb{C}^{N_S \times N_S}$ is the MIMO transfer matrix and $\mathbf{n}_\mathbf{G}$ represents spatially correlated Gaussian noise. It is evident that, whenever the transfer matrix is not diagonal, the data symbols of different

streams interfere with each other. The design of the precoding and combining matrices is therefore essential to suppress this inter-stream interference and improve overall system performance.

2.2.2 MU-MIMO System Model

When multiple receivers are served simultaneously by a single multi-antenna transmitter, the system is referred to as a multi-user (MU) MIMO system. In this context, the transmitter is called the base station (BS) and the individual receivers are referred to as user terminals (UTs). A block diagram is shown in Fig. 2.3.

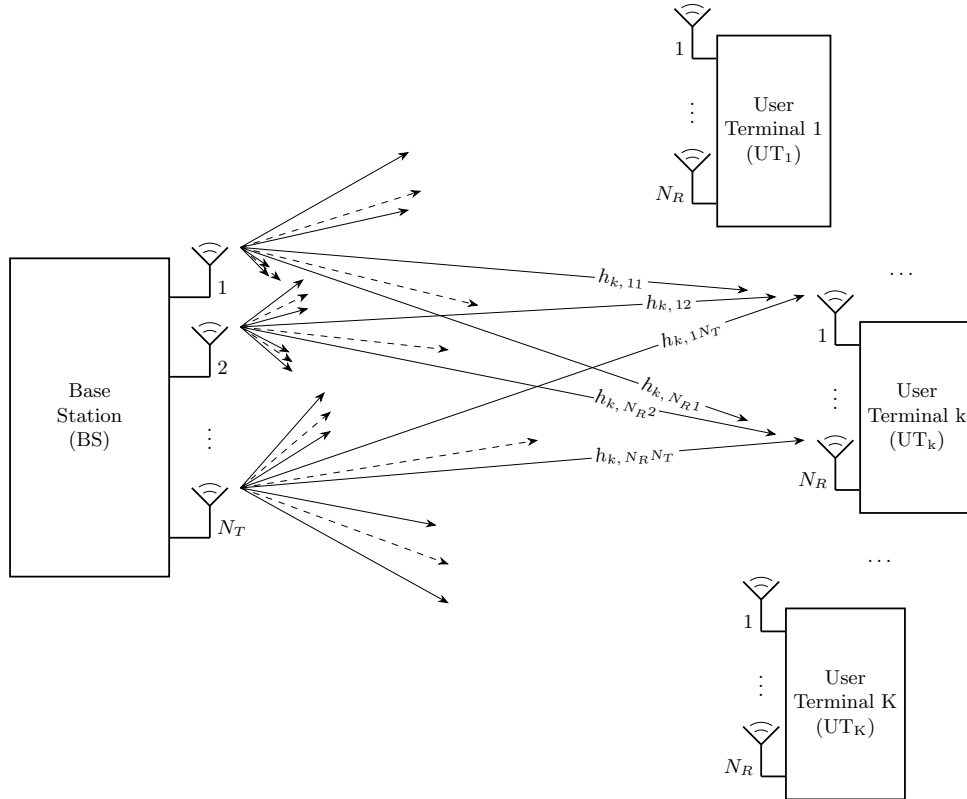


Figure 2.3: Block diagram of a MU-MIMO system.

Throughout this work, the BS is assumed to be equipped with more antennas than all UTs combined, and each UT is assumed to receive at most as many data streams as it has receive antennas. For notational convenience, all UTs are assumed to receive the same number of data streams, denoted by N_S . In practical scenarios where this does not hold, N_S is defined as the maximum number of data streams intended for any single UT. The data symbol vectors intended for a UT receiving fewer streams are padded with zeros.

Analogous to the SU-MIMO case, the communication between the BS and a certain UT_k at a particular time instant is characterized by

$$\mathbf{y}_k = \mathbf{H}_k \mathbf{x} + \mathbf{n}_k, \quad k = 1, \dots, K, \quad (2.6)$$

where $\mathbf{H}_k \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix that describes the link between the BS and UT_k .

The transmitted signal vector $\mathbf{x} \in \mathbb{C}^{N_T}$ is now a function of the data symbol vectors intended for different UTs. The BS assigns a dedicated precoding scheme $\mathbf{F}_k \in \mathbb{C}^{N_T \times N_S}$ to the data symbol vector $\mathbf{a}_k$ intended for UT_k, and transmits the superposition of all precoded streams:

$$\mathbf{x} = \sum_{k=1}^K \mathbf{F}_k \mathbf{a}_k. \quad (2.7)$$

Since the data symbols intended for different UTs are assumed to be uncorrelated, the total transmit power constraint can now be formulated as (see Appendix E)

$$\mathbb{E}[\|\mathbf{x}\|^2] = \sum_{k=1}^K \text{tr}(\mathbf{F}_k^H \mathbf{F}_k) \leq P_t. \quad (2.8)$$

Applying the combining scheme $\mathbf{G}_k \in \mathbb{C}^{N_S \times N_R}$ at UT_k to the received signal vector and substituting (2.7) into (2.6) yields

$$\mathbf{z}_k = \underbrace{(\mathbf{G}_k \mathbf{H}_k \mathbf{F}_k) \mathbf{a}_k}_{\text{desired signal}} + \underbrace{\sum_{k'=1, k' \neq k}^K (\mathbf{G}_k \mathbf{H}_k \mathbf{F}_{k'}) \mathbf{a}_{k'}}_{\text{inter-user interference}} + \underbrace{\mathbf{G}_k \mathbf{n}_k}_{\text{noise}}. \quad (2.9)$$

The key difference from the SU-MIMO scenario is the presence of inter-user interference. Each UT receives not only its desired signal but also the signals intended for all other UTs. Suppressing this interference through precoder design at the BS is an important challenge in MU-MIMO systems.

The overall input-output relation of the MU-MIMO Gaussian channel can be written compactly as

$$\mathbf{z} = (\mathbf{G} \mathbf{H} \mathbf{F}) \mathbf{a} + \mathbf{G} \mathbf{n}, \quad \text{with} \quad (2.10)$$

$$\mathbf{z} = \begin{bmatrix} \mathbf{z}_1 \\ \vdots \\ \mathbf{z}_K \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \mathbf{G}_1 & & \\ & \ddots & \\ & & \mathbf{G}_K \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_1 \\ \vdots \\ \mathbf{H}_K \end{bmatrix}, \quad \mathbf{F} = [\mathbf{F}_1 \quad \dots \quad \mathbf{F}_K], \quad \mathbf{a} = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_K \end{bmatrix}.$$

where $\mathbf{G} \in \mathbb{C}^{KN_S \times KN_R}$ denotes the compound combining matrix, $\mathbf{H} \in \mathbb{C}^{KN_R \times N_T}$ the compound channel matrix, and $\mathbf{F} \in \mathbb{C}^{N_T \times KN_S}$ the compound precoding matrix.

2.3 Information Theory Concepts

In this section, we introduce the information-theoretic concepts needed for the analysis of MIMO communication systems. We first discuss mutual information, then define the achievable rate and capacity for SU-MIMO channels, and finally extend these concepts to the MU scenario.

2.3.1 Mutual Information

Consider a single use of a SU-MIMO Gaussian channel, as described by (2.1). The transmitter generates a random vector $\mathbf{x} \in \mathbb{C}^{N_T}$ according to the input distribution with probability density function (pdf) $p(\mathbf{x})$, while the receiver observes the random vector $\mathbf{y} \in \mathbb{C}^{N_R}$. The relation between the channel input $\mathbf{x}$ and the channel output $\mathbf{y}$ is probabilistic and is fully characterized by the channel law, described by the conditional pdf $p(\mathbf{y} | \mathbf{x})$.

The number of information bits transmitted per channel use is referred to as the information bit rate (IBR). The mutual information (MI) between $\mathbf{x}$ and $\mathbf{y}$ quantifies how much information the received vector $\mathbf{y}$ provides about the transmitted vector $\mathbf{x}$. In other words, it represents the average reduction in the uncertainty about $\mathbf{x}$ obtained from observing $\mathbf{y}$. It is given by

$$I(\mathbf{x}; \mathbf{y}) = \mathbb{E} \left[\log_2 \left(\frac{p(\mathbf{y} | \mathbf{x})}{p(\mathbf{y})} \right) \right], \quad (2.11)$$

where the expectation is taken with respect to the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ and $p(\mathbf{y})$ is the marginal distribution of $\mathbf{y}$.

According to Shannon's channel coding theorem [3], reliable communication is possible if the transmission IBR is strictly smaller than the MI between $\mathbf{x}$ and $\mathbf{y}$. Here, reliable communication means that the decoding error probability can be made arbitrarily small by using sufficiently long channel codes.

2.3.2 Achievable Rate and Capacity

For a given input distribution $p(\mathbf{x})$ and channel law $p(\mathbf{y} | \mathbf{x})$, any IBR r smaller than the MI is said to be achievable. The MI is thus a supremum (least upper bound) on the achievable rate.

The capacity C of a SU-MIMO channel is the largest achievable rate. Equivalently, it is the highest spectral efficiency for which reliable communication is possible. It is defined as the maximum MI between $\mathbf{x}$ and $\mathbf{y}$ over all possible input distributions $p(\mathbf{x})$:

$$C = \max_{p(\mathbf{x})} I(\mathbf{x}; \mathbf{y}). \quad (2.12)$$

Typically, the maximization is subject to the transmit power constraint, as formulated in (2.3).

In [4], the capacity for a SU-MIMO Gaussian communication system is derived. First, it is shown that under the total transmit power constraint, the MI is maximized in case of Gaussian signaling, i.e. the channel input $\mathbf{x}$ is CS complex Gaussian distributed. Then, the mutual information between $\mathbf{x}$ and $\mathbf{y}$ for a given covariance matrix $\mathbf{C}_x$ is obtained as

$$I(\mathbf{x}; \mathbf{y}) = \log_2 \left(\det \left(\mathbf{I}_{N_R} + \mathbf{C}_n^{-1} \mathbf{H} \mathbf{C}_x \mathbf{H}^H \right) \right). \quad (2.13)$$

The capacity then results from maximizing (2.13) over all positive semidefinite covariance matrices $\mathbf{C}_x$ that satisfy the transmit power constraint:

$$\begin{aligned} C &= \max_{\mathbf{C}_x} \log_2 (\det (\mathbf{I}_{N_R} + \mathbf{C}_n^{-1} \mathbf{H} \mathbf{C}_x \mathbf{H}^H)) \\ \text{s.t. } &\mathbf{C}_x \succeq 0, \quad \text{tr}(\mathbf{C}_x) \leq P_t. \end{aligned} \quad (2.14)$$

2.3.3 Rate Region and Sum Capacity

In contrast to the SU scenario, where the IBR is a scalar r , a MU-MIMO channel is characterized by a rate tuple $(r_1, \dots, r_K)$, where each element r_k denotes the IBR of a single UT. The rate region $\mathcal{R}$ is the set of all rate tuples whose individual IBRs are simultaneously achievable under the given system constraints. The rate region is a convex set, since any convex combination of achievable rate tuples is itself achievable through time-sharing between the corresponding transmission strategies. An example rate region for $K = 2$ users is shown in Fig. 2.4.

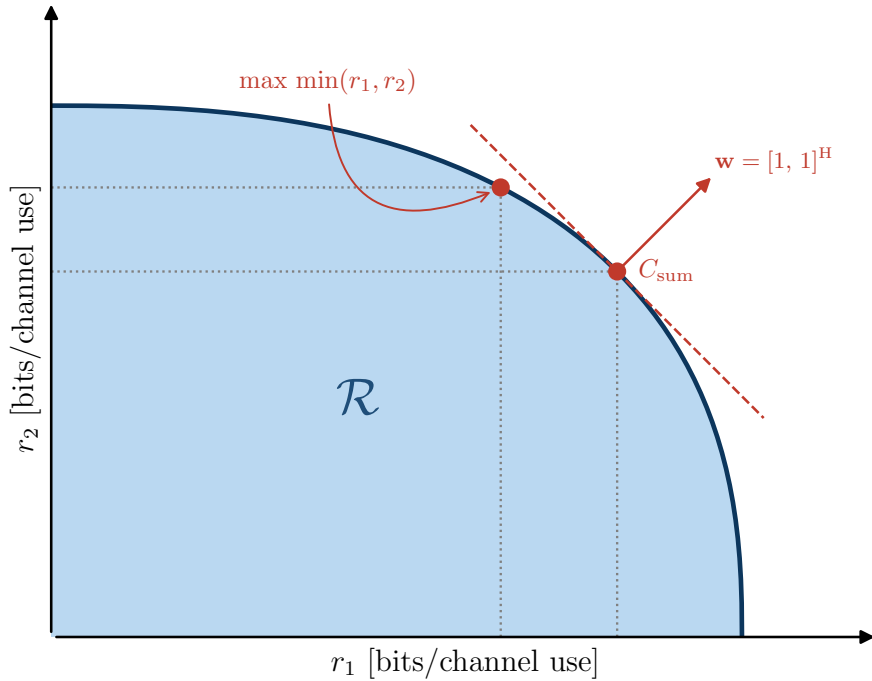


Figure 2.4: Example of a rate region $\mathcal{R}$ for a MU-MIMO system with $K = 2$ users.

The sum capacity point is the boundary point whose outward normal is the all-ones vector. The max-min point is given by the intersection of the boundary with the line $r_1 = r_2$. It corresponds to a fair operating point where both users achieve the same IBR.

The rate region captures the trade-offs inherent to multi-user transmission: increasing the IBR of one user may reduce the rates achievable by others due to interference and shared resources. It is therefore desirable to operate on the boundary of the rate region, where no user's rate can be increased without reducing that of another.

The sum capacity C_{sum} is the maximum achievable sum rate (SR), i.e., the sum of all individual IBRs:

$$C_{\text{sum}} = \max_{(r_1, \dots, r_K) \in \mathcal{R}} \sum_{k=1}^K r_k. \quad (2.15)$$

More generally, we define the weighted sum rate (WSR) as

$$\text{WSR} = \sum_{k=1}^K w_k r_k, \quad (2.16)$$

where $w_k \geq 0$ is the weight assigned to UT_k . The full boundary of the rate region can then be traced by maximizing the WSR over all achievable rate tuples for varying weight vectors $\mathbf{w} = (w_1, \dots, w_K)$. The sum capacity corresponds to the special case where all weights are equal to one. The maximization in both cases is typically subject to the total transmit power constraint (2.8).

In [5, 6, 7], the sum capacity of the MU-MIMO Gaussian channel is derived by exploiting the uplink–downlink duality: the downlink broadcast channel and the uplink multiple-access channel yield the same rate region under the same total transmit power constraint. It therefore suffices to solve the uplink problem, which is more tractable.

In the uplink channel, all K UTs transmit simultaneously, with UT_k sending a vector $\mathbf{x}_k^{\text{up}} \in \mathbb{C}^{N_R}$ to the BS. The MI between the transmitted vectors $\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}}$ and the received vector $\mathbf{y}$, given input covariance matrices $\mathbf{C}_{\mathbf{x},k}$, is defined as

$$I(\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}}; \mathbf{y}) = \mathbb{E} \left[\log_2 \left(\frac{p(\mathbf{y} | \mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}})}{p(\mathbf{y})} \right) \right], \quad (2.17)$$

where the expectation is taken with respect to the joint pdf of $(\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}})$ and $\mathbf{y}$, and $p(\mathbf{y})$ is the marginal pdf of $\mathbf{y}$. It is further obtained as

$$I(\mathbf{x}_1^{\text{up}}, \dots, \mathbf{x}_K^{\text{up}}; \mathbf{y}) = \log_2 \left(\det \left(\mathbf{I}_{N_T} + \mathbf{C}_n^{-1} \sum_{k=1}^K \mathbf{H}_k \mathbf{C}_{\mathbf{x},k}^{\text{up}} \mathbf{H}_k^H \right) \right). \quad (2.18)$$

The uplink SR is achievable if and only if the sum of the IBRs of all UTs is strictly smaller than the MI as given in (2.18). Therefore, the sum capacity of the downlink channel is obtained by maximizing (2.18) over all positive semidefinite covariance matrices satisfying the total transmit power constraint and applying the duality result:

$$\begin{aligned} C_{\text{sum}} &= \max_{\mathbf{C}_{\mathbf{x},1}^{\text{up}}, \dots, \mathbf{C}_{\mathbf{x},K}^{\text{up}}} \log_2 \left(\det \left(\mathbf{I}_{N_T} + \mathbf{C}_n^{-1} \sum_{k=1}^K \mathbf{H}_k \mathbf{C}_{\mathbf{x},k}^{\text{up}} \mathbf{H}_k^H \right) \right) \\ \text{s.t.} \quad &\sum_{k=1}^K \text{tr}(\mathbf{C}_{\mathbf{x},k}^{\text{up}}) \leq P_t \quad \text{and} \quad \mathbf{C}_{\mathbf{x},k}^{\text{up}} \succeq 0 \quad \forall k = 1, \dots, K. \end{aligned} \quad (2.19)$$

Chapter 3

Precoding for SU-MIMO systems

This chapter develops linear precoding strategies for SU-MIMO systems under a total transmit-power constraint. First, we analyze the capacity in greater detail. Next, a generalized WMMSE design framework for the joint optimization of the precoder and combiner is derived. Finally, the performance is evaluated through Monte-Carlo simulations for different system configurations.

3.1 Capacity Achieving Precoding

Consider the SU-MIMO system model introduced in Section 2.2.1 together with the associated capacity optimization problem in (2.14). We first derive a more explicit expression for the channel capacity and then show that this capacity can be achieved by means of linear precoding. Parts of this section follow a structure similar to that in Chapter 3 of [8].

3.1.1 SU-MIMO Capacity

The singular value decomposition (SVD) [9] of the channel matrix yields

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H, \quad (3.1)$$

where $\mathbf{U} \in \mathbb{C}^{N_R \times N_R}$ and $\mathbf{V} \in \mathbb{C}^{N_T \times N_T}$ are unitary matrices, and $\mathbf{\Sigma} \in \mathbb{R}^{N_R \times N_T}$ is a rectangular diagonal matrix which contains the singular values of $\mathbf{H}$ on its main diagonal in decreasing order. The columns of $\mathbf{U}$ and $\mathbf{V}$ contain the left and right singular vectors of $\mathbf{H}$, respectively.

Now define the transformed variables $\tilde{\mathbf{y}} = \mathbf{U}^H \mathbf{y}$, $\tilde{\mathbf{x}} = \mathbf{V}^H \mathbf{x}$, and $\tilde{\mathbf{n}} = \mathbf{U}^H \mathbf{n}$. Since multiplication by a unitary matrix preserves the distribution of CS complex Gaussian noise, $\tilde{\mathbf{n}}$ has the same distribution as $\mathbf{n}$. Moreover, because $\mathbf{U}$ and $\mathbf{V}$ are unitary, the transformations between $(\mathbf{x}, \mathbf{y})$ and $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ are bijective. The original channel from (2.1) can therefore be represented equivalently as

$$\tilde{\mathbf{y}} = \mathbf{\Sigma} \tilde{\mathbf{x}} + \tilde{\mathbf{n}}, \quad (3.2)$$

which is obtained by applying the SVD of the channel matrix and performing a change of variables (see Appendix E). The channel in (3.2) is equivalent to the original channel in the sense that the

transformation preserves the mutual information between input and output. Consequently, both channels have the same capacity.

The channel matrix $\mathbf{H}$ has exactly as many non-zero singular values $\{\sigma_i\}$ as its rank, denoted as R_H . Written component-wise, the equivalent channel becomes

$$\tilde{y}_i = \sigma_i \tilde{x}_i + \tilde{n}_i \quad \text{for } i = 1, \dots, R_H, \quad (3.3)$$

with $\tilde{y}_i = \tilde{n}_i$ for $R_H < i \leq N_R$. The equivalent channel decomposes into R_H independent parallel scalar Gaussian subchannels, each with its own channel gain σ_i . These subchannels and their gains are referred to as eigen subchannels and eigenmodes, respectively. Since the components $\tilde{x}_i$ for $R_H < i \leq N_T$ have no effect on the output, they carry no information and can be set to zero.

For each eigen subchannel, the mutual information (MI) is maximised by a zero-mean, CS complex Gaussian input [4]. Under this choice, the optimization problem (2.14) simplifies to

$$\begin{aligned} C &= \max_{\{\mathbb{E}[|\tilde{x}_i|^2]\}} \sum_{i=1}^{R_H} \log_2 \left(1 + \mathbb{E}[|\tilde{x}_i|^2] \frac{\sigma_i^2}{N_0} \right) \\ \text{s.t. } &\sum_{i=1}^{R_H} \mathbb{E}[|\tilde{x}_i|^2] \leq P_t. \end{aligned} \quad (3.4)$$

This maximization problem is solved using the Lagrange multiplier method (see Appendix A). The variances of the channel input need to be chosen according to the waterfilling solution [10]:

$$\mathbb{E}[|\tilde{x}_i|^2] = \left(\mu_w - \frac{N_0}{\sigma_i^2} \right)_+, \quad (3.5)$$

where the water level μ_w is found iteratively to satisfy the transmit power constraint with equality.

The resulting capacity is given by

$$C = \sum_{i=1}^{R_H} \log_2 \left(\max(1, \mu_w \frac{\sigma_i^2}{N_0}) \right). \quad (3.6)$$

3.1.2 Optimal Precoder Design

In this section, we design a linear precoder and combiner that together achieve the capacity of the SU-MIMO channel. We assume that the data symbol vector is CS complex Gaussian distributed and spatially uncorrelated, and restrict ourselves to the case where the number of data streams is not fixed in advance, but can be chosen freely as part of the design.

For an arbitrary precoding matrix $\mathbf{F}$ and combining matrix $\mathbf{G}$, the data processing inequality [11] yields $I(\mathbf{x}; \mathbf{y}) \geq I(\mathbf{a}; \mathbf{z})$, where the MI between the transmitted data symbols $\mathbf{a}$ and the decision

variables $\mathbf{z}$ is given by:

$$I(\mathbf{a}; \mathbf{z}) = \log_2(\det(\mathbf{I}_{N_s} + (\mathbf{G} \mathbf{C}_n \mathbf{G}^H)^{-1} (\mathbf{G} \mathbf{H} \mathbf{F}) \mathbf{C}_a (\mathbf{G} \mathbf{H} \mathbf{F})^H)) \quad (3.7a)$$

$$= \log_2\left(\det\left(\mathbf{I}_{N_s} + \frac{1}{N_0}(\mathbf{G} \mathbf{G}^H)^{-1} (\mathbf{G} \mathbf{H} \mathbf{F}) (\mathbf{G} \mathbf{H} \mathbf{F})^H\right)\right). \quad (3.7b)$$

Expression (3.7a) follows directly from (2.13) and the structural similarity between (2.1) and (2.5). Expression (3.7b) then follows from (3.7a) upon substituting $\mathbf{C}_a = \mathbf{I}_{N_s}$ and $\mathbf{C}_n = N_0 \mathbf{I}_{N_R}$.

Motivated by the capacity derivation in the previous subsection, we choose the columns of $\mathbf{F}$ and the rows of $\mathbf{G}$ as the R_H most dominant right and left singular vectors of $\mathbf{H}$, respectively. These are precisely the singular vectors associated with the non-zero singular values of $\mathbf{H}$. The number of data streams is therefore chosen equal to the rank of the channel matrix. In addition, the precoder incorporates the power allocation through a diagonal matrix $\mathbf{P}$. More specifically, we set

$$\mathbf{F} = \mathbf{V}_{N_s} \sqrt{\mathbf{P}} \quad \text{and} \quad \mathbf{G} = \mathbf{U}_{N_s}^H. \quad (3.8)$$

It follows from (3.8) that $I(\mathbf{a}; \mathbf{z}) = I(\mathbf{x}; \mathbf{y})$ for an appropriate choice of the power allocation matrix $\mathbf{P}$. This result can be verified in a brute-force way by substituting the particular precoder and combiner matrices from (3.8) into (3.7) and replacing the channel matrix by its SVD (see Appendix E).

An alternative and more elegant approach consists in substituting the particular precoder and combiner into (2.5) and comparing the resulting expression with the equivalent channel given in (3.2). This yields

$$\mathbf{z} = \mathbf{U}_{N_s}^H \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H \mathbf{V}_{N_s} \sqrt{\mathbf{P}} \mathbf{a} + \mathbf{U}_{N_s}^H \mathbf{n} \quad (3.9a)$$

$$= \mathbf{\Sigma}_{N_s} \sqrt{\mathbf{P}} \mathbf{a} + \tilde{\mathbf{n}}, \quad (3.9b)$$

where $\mathbf{\Sigma}_{N_s} \in \mathbb{R}^{N_s \times N_s}$ contains the R_H non-zero singular values of $\mathbf{H}$ on its diagonal. Comparing (3.9) with (3.2), one can readily verify that $I(\mathbf{a}; \mathbf{z}) = I(\mathbf{x}; \mathbf{y})$, provided that the power allocated to each stream is set according to the waterfilling solution. Hence, transmission with the precoder and combiner from (3.8) achieves the capacity of the SU-MIMO channel.

Denoting the s th element of the decision variable vector $\mathbf{z}$ by z_s , one can write (3.9) component-wise as

$$z_s = \sigma_s \sqrt{p_s} a_s + \tilde{n}_s \quad \text{for } s = 1, \dots, N_s. \quad (3.10)$$

This expression shows that the precoder and combiner in (3.8) diagonalize the SU-MIMO transfer matrix from (2.5). The original SU-MIMO channel is decoupled into $N_s = R_H$ independent parallel scalar single-input single-output (SISO) Gaussian subchannels with gains σ_s , which coincides exactly with the decomposition of the equivalent channel in (3.3). A visualization of this decomposition is shown in Fig. 3.1.

As a result, the individual data streams become statistically independent. Detection therefore reduces to symbol-by-symbol processing rather than joint detection of the full decision-variable

vector. This leads to a significant reduction in receiver complexity, especially for systems with a large number of antennas and data streams.

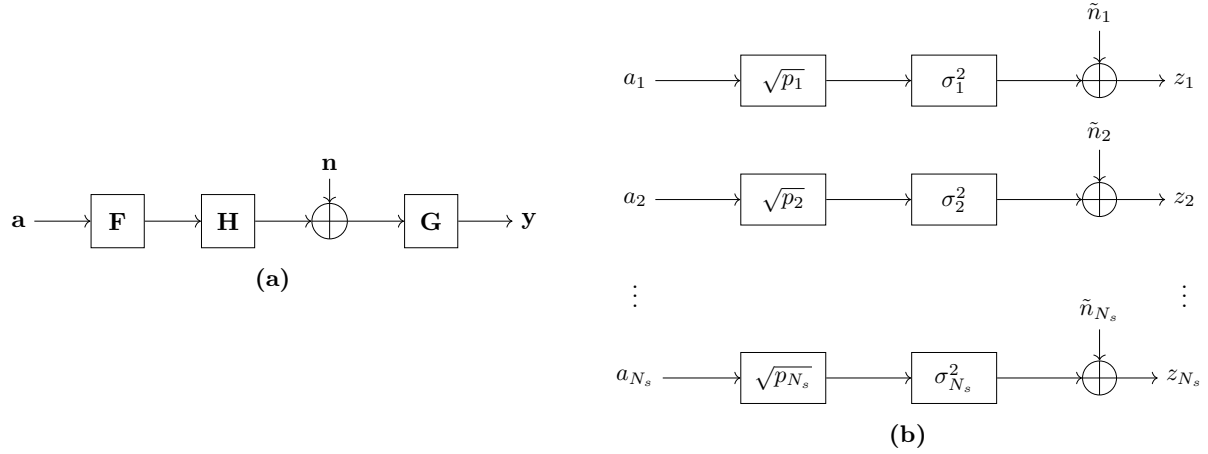


Figure 3.1: A block diagram of the original SU-MIMO channel (a) and its decomposition into parallel SISO Gaussian eigen subchannels (b) by means of the optimal precoder and combiner.

No general closed-form expression exists for the optimal water level μ_w , and hence for the resulting power allocation. Nevertheless, the optimal power allocation is fully characterized by

$$\sum_{i=1}^{N_s} p_i = P_t \quad \text{and} \quad p_i = \left(\mu_w - \frac{N_0}{\sigma_i^2} \right)_+ . \quad (3.11)$$

Since the total allocated power is a monotonically increasing function of the water level, the latter can be determined numerically by means of a bisection method. A more efficient approach exploits the geometric interpretation of the waterfilling solution. In the present SU-MIMO setting, each eigen subchannel can be represented as a container with unit width and floor level N_0/σ_i^2 , while the total transmit power is interpreted as water that is poured into these containers until a common water level is reached. The complete procedure is summarized in Appendix A.2. The structure of the optimal power allocation is illustrated in Fig. 3.2 for a SU-MIMO system with four transmit and four receive antennas operating at an signal-to-noise ratio (SNR) of 5 dB: eigen subchannels with stronger channel conditions, i.e. larger gains σ_i , receive more power than weaker ones.

At extremely low SNR, the waterfilling solution corresponds to the eigenbeamforming power allocation strategy, which concentrates all available power on the strongest eigen subchannel. As a result, only a single data stream is transmitted in this SNR regime. This behavior is illustrated in Fig. 3.3a. Conversely, at an extremely high SNR, the optimal power allocation strategy reduces to the uniform power allocation, which distributes power equally across all eigen subchannels, as illustrated in Fig. 3.3b. It is worth noting that uniform power allocation does not require knowledge of the current channel matrix, in contrast to the general waterfilling solution.

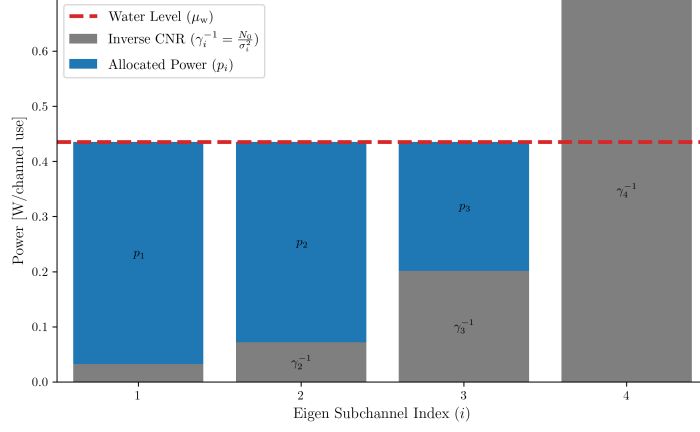


Figure 3.2: Optimal power allocation for a SU-MIMO system with 4 transmit and 4 receive antennas and a total transmit power P_t of 1 W per channel use. The system operates at an SNR of 5 dB.

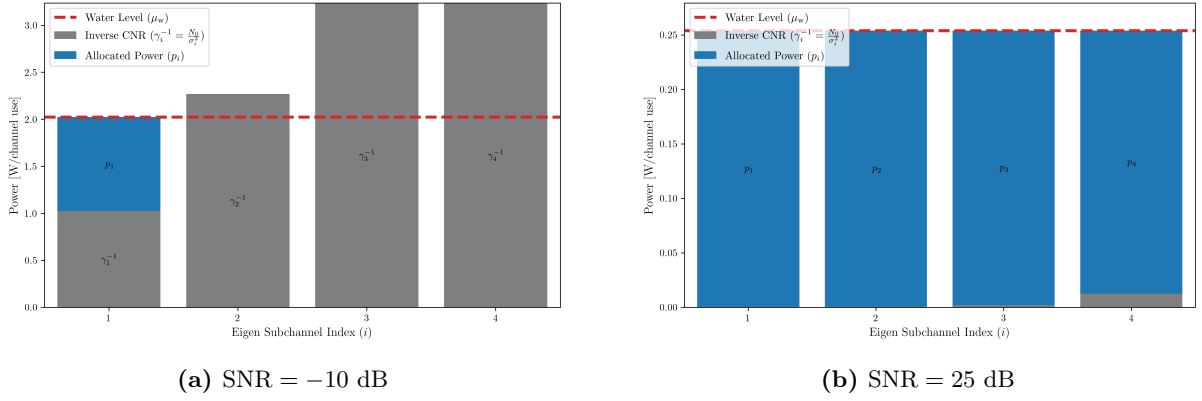


Figure 3.3: Waterfilling power allocation for a SU-MIMO system with 4 transmit and 4 receive antennas at different SNR extremes. The maximum transmit power P_t is 1 W per channel use.

3.2 Joint Design for Linear Precoder and Combiner

In the previous section, a linear precoder and combiner that jointly decompose the SU-MIMO channel into independent parallel eigen subchannels were shown to achieve the channel capacity. In this section, we generalize this result by adopting the minimization of a weighted sum of mean-square estimation errors as the design criterion. It will be shown that the optimum precoder and combiner under this criterion likewise decouple the channel into parallel eigen subchannels, regardless of the specific choice of error weights. By an appropriate selection of these weights, the framework specializes to the maximum information rate design, the Quality-of-Service (QoS) design, and the unweighted minimum mean squared error (MMSE) design.

3.2.1 Generalized WMMSE Design

The objective is to jointly design the precoder $\mathbf{F}$ and the combiner $\mathbf{G}$ so as to minimize a wweighted combination of mean-square estimation errors. To this end, the estimation error vector $\mathbf{e} \in \mathbb{C}^{N_s}$ is defined as the difference between the transmitted data symbol vector $\mathbf{a}$ and the decision variable vector $\mathbf{z}$:

$$\mathbf{e} = \mathbf{a} - \mathbf{z} = \mathbf{a} - (\mathbf{G}\mathbf{H}\mathbf{F}\mathbf{a} + \mathbf{G}\mathbf{n}). \quad (3.12)$$

The mean squared error (MSE) matrix, also referred to as the error covariance matrix, is defined as

$$\text{MSE} = \mathbb{E}[\mathbf{e}\mathbf{e}^H] = (\mathbf{G}\mathbf{H}\mathbf{F} - \mathbf{I}_{N_s})(\mathbf{G}\mathbf{H}\mathbf{F} - \mathbf{I}_{N_s})^H + \mathbf{G}\mathbf{C}_n\mathbf{G}^H \quad (3.13)$$

where the last equality follows from the independence of the data symbols $\mathbf{a}$ and the noise vector $\mathbf{n}$, and the assumption that the data symbols are spatially uncorrelated and normalized.

The weighted mean squared error (WMSE) is defined as

$$\text{WMSE} = \mathbb{E}\left[\left\|\mathbf{W}_e^{1/2}\mathbf{e}\right\|^2\right] = \mathbb{E}\left[\text{tr}(\mathbf{W}_e^{1/2}\mathbf{e}\mathbf{e}^H\mathbf{W}_e^{1/2H})\right] = \text{tr}(\mathbf{W}_e\mathbb{E}[\mathbf{e}\mathbf{e}^H]), \quad (3.14)$$

where $\mathbf{W}_e$ is a diagonal positive definite weight matrix that assigns a relative importance to the error of each data stream.

The weighted minimum mean squared error (WMMSE) optimization problem is then formulated as

$$\begin{aligned} \min_{\mathbf{F}, \mathbf{G}} \quad & \text{tr}(\mathbf{W}_e\mathbb{E}[\mathbf{e}\mathbf{e}^H]) \\ \text{s. t.} \quad & \text{tr}(\mathbf{F}\mathbf{F}^H) \leq P_t. \end{aligned} \quad (3.15)$$

The optimization problem in (3.15) can be solved using the Lagrange duality together with the Karush-Kuhn-Tucker (KKT) conditions. This leads to the following necessary optimality conditions [12]:

$$\mathbf{H}\mathbf{F} = \mathbf{H}\mathbf{F}(\mathbf{G}\mathbf{H}\mathbf{F})^H + \mathbf{C}_n\mathbf{G}^H \quad (3.16a)$$

$$\mathbf{W}_e\mathbf{G}\mathbf{H} = (\mathbf{G}\mathbf{H}\mathbf{F})^H\mathbf{W}_e\mathbf{G}\mathbf{H} + \mu\mathbf{F}^H \quad (3.16b)$$

where $\mu > 0$ is the Lagrange multiplier associated with the total transmit power constraint.

Solving these coupled matrix equations yields the optimum linear precoder and combiner. As shown in [12], they admit the following closed-form expressions:

$$\mathbf{F} = \mathbf{V}\Phi_f \quad (3.17a)$$

$$\mathbf{G} = \Phi_g\mathbf{V}^H\mathbf{H}^H\mathbf{C}_n^{-1} \quad (3.17b)$$

The matrices $\Phi_f, \Phi_g \in \mathbb{R}^{R_H \times R_H}$ are diagonal matrices with non-negative diagonal entries, and are given by

$$\Phi_f = (\mu^{-1/2}\mathbf{\Lambda}^{-1/2}\mathbf{W}_e^{1/2} - \mathbf{\Lambda}^{-1})_+^{1/2} \quad (3.18a)$$

$$\Phi_g = (\mu^{1/2}\mathbf{\Lambda}^{-1/2}\mathbf{W}_e^{-1/2} - \mu\mathbf{\Lambda}^{-1}\mathbf{W}_e^{-1})_+^{1/2}\mathbf{\Lambda}^{1/2} \quad (3.18b)$$

The matrices $\mathbf{V} \in \mathbb{C}^{N_T \times R_H}$ and $\mathbf{\Lambda} \in \mathbb{R}^{R_H \times R_H}$ are obtained from the eigenvalue decomposition (EVD) of the noise-whitened Gram matrix of $\mathbf{H}$:

$$\mathbf{H}^H\mathbf{C}_n^{-1}\mathbf{H} = [\mathbf{V} \quad \bar{\mathbf{V}}] \begin{bmatrix} \mathbf{\Lambda} & \mathbf{0} \\ \mathbf{0} & \bar{\mathbf{\Lambda}} \end{bmatrix} [\mathbf{V} \quad \bar{\mathbf{V}}]^H, \quad (3.19)$$

where the columns of $\mathbf{V}$ form an orthonormal basis for the range space of $(\mathbf{H}^H \mathbf{C}_n^{-1} \mathbf{H})$, and the columns of $\bar{\mathbf{V}}$ span its null space. The diagonal matrix $\mathbf{\Lambda}$ contains the R_H non-zero eigenvalues arranged in descending order, whereas $\bar{\mathbf{\Lambda}}$ contains the remaining zero eigenvalues. Note that, as in Section 3.1.2, the number of data streams is set equal to the rank of the channel matrix, i.e. $N_s = R_H$. Moreover, the rank of the channel matrix is equal to the rank of $(\mathbf{H}^H \mathbf{C}_n^{-1} \mathbf{H})$ since $\mathbf{C}_n^{-1}$ is positive definite and therefore does not alter the rank of the product.

Now, it only remains to show how to obtain the Lagrange multiplier that satisfies the power constraint and also is such that the bracketed matrices in both (3.18a) and (3.18b) are positive-definite. Let $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_{R_H})$ and $\mathbf{W}_e = \text{diag}(w_1, \dots, w_{R_H})$. Suppose $k \leq R_H$ eigen subchannels are effectively used for transmission, i.e. they are allocated non-zero power. From (3.18a) and the total power constraint in (3.15), we have

$$\text{tr}(\mathbf{F}\mathbf{F}^H) = \text{tr}(\mathbf{\Phi}_f^2) = \mu^{-1/2} \sum_{i=1}^k \lambda_i^{-1/2} w_i^{1/2} - \sum_{i=1}^k \lambda_i^{-1} = P_t. \quad (3.20)$$

Therefore, we can obtain the following expression for the Lagrange multiplier μ :

$$\mu = \left(\frac{\sum_{i=1}^k \lambda_i^{-1/2} w_i^{1/2}}{P_t + \sum_{i=1}^k \lambda_i^{-1}} \right)^2. \quad (3.21)$$

Let $\rho_k = \lambda_k w_k$ and let them be ordered in decreasing manner. To compute $\mathbf{\Phi}_f$ optimally and ensure that it is positive-definite, we can compute μ according to (3.21) for each $k = 1, \dots, R_H$ iteratively, and select the largest k such that $\mu \leq \rho_k$ [12].

From (3.17a) and (3.17b), we have:

$$\mathbf{T} = \mathbf{G}\mathbf{H}\mathbf{F} = \mathbf{\Phi}_g \mathbf{\Lambda} \mathbf{\Phi}_f. \quad (3.22)$$

In other words, the optimum precoder and combiner diagonalize the SU-MIMO transfer matrix $\mathbf{T}$, and hence decouple the channel into parallel eigen subchannels, regardless of the specific choice of error weights. This result is similar to the capacity-achieving precoder and combiner derived in Section 3.1.2, which likewise decompose the channel matrix into parallel eigen subchannels.

By choosing the error weight matrix appropriately, the generalized WMMSE framework can be specialized to different design criteria, such as maximum information rate, unweighted MMSE, or prescribed quality-of-service requirements. The following subsections discuss these special cases in more detail.

3.2.2 Maximum Information Rate Design

It was shown in Section 3.1.2 that the capacity of the SU-MIMO channel can be achieved by a linear precoder and combiner that diagonalize the transfer matrix and allocate power according to the waterfilling solution. Rewriting (3.11) in matrix format and switching to the parameter notation of the generalized design yields [12]

$$\mathbf{\Phi}_f = (\mu^{-1/2} \mathbf{I} - \mathbf{\Lambda}^{-1})_+^{1/2}. \quad (3.23)$$

The above equation is obtained by choosing the error weight matrix as $\mathbf{W}_e = \mathbf{\Lambda}$ in (3.18a). It is shown in [12] that, for this particular choice of error weights, the closed-form solutions in (3.17a) and (3.17b) yield the linear precoder and combiner that indeed maximize the information rate. In other words, the maximum information rate design is just a special case of the generalized WMMSE design. The choice of the error weights $\mathbf{W}_e = \mathbf{\Lambda}$ indicates that stronger eigen subchannels support higher rates compared to the weaker ones, and hence need to be more heavily weighted in the cost function. This is consistent with the waterfilling solution, which allocates more power to stronger subchannels.

The maximum information rate design is particularly relevant in adaptive modulation systems, where more power and higher order modulation are used on eigen subchannels with higher gains to improve the total data rate. The performance of these systems will be evaluated in Section ??, based on simulation results for different system configurations.

3.2.3 Quality of Service Design

Consider an application in which different data streams carry information of heterogeneous types with different MSE requirements for successful transmission. A typical example is a multimedia system in which audio and video streams are transmitted simultaneously but require different levels of reliability. For such Quality-of-Service (QoS)-sensitive applications, the system must be designed to achieve prescribed MSE levels on the individual eigen subchannels.

First, we note that the expression for the MSE matrix simplifies upon substituting the optimum precoder and combiner from (3.17a) and (3.17b):

$$\text{MSE} = \mathbb{E}[\mathbf{e} \mathbf{e}^H] = \mathbf{W}_e^{-1/2} \mathbf{\Lambda}^{-1/2} \mu^{1/2}. \quad (3.24)$$

Next, define the output SNR matrix as

$$\text{SNR} = (\mathbf{G} \mathbf{H} \mathbf{F})^H (\mathbf{G} \mathbf{C}_n \mathbf{G}^H)^{-1} (\mathbf{G} \mathbf{H} \mathbf{F}) \quad (3.25a)$$

$$= \mathbf{\Phi}_f^2 \mathbf{\Lambda} = (\mu^{-1/2} \mathbf{\Lambda}^{1/2} \mathbf{W}_e^{1/2} - \mathbf{I})_+, \quad (3.25b)$$

where we again used the expressions for the optimum precoder and combiner to simplify the first equality. The relation $\text{SNR} = (\text{MSE}^{-1} - \mathbf{I})_+$ follows directly from these expressions, showing that the SNR and MSE on each subchannel are in a one-to-one inverse correspondence. Hence, prescribing relative MSEs is equivalent to prescribing relative output SNRs. In particular, equal SNRs across all eigen subchannels imply equal MSEs, regardless of the specific choice of error weights.

We now determine the error weight matrix $\mathbf{W}_e$ required to realize a desired set of relative output SNRs across the eigen subchannels. To this end, let $\mathbf{D}$ be a diagonal matrix containing the desired relative SNRs, and let $d > 0$ denote an arbitrary scaling factor. The design requirement reduces to the following equation

$$d \mathbf{D} = (\mu^{-1/2} \mathbf{\Lambda}^{1/2} \mathbf{W}_e^{1/2} - \mathbf{I})_+, \quad (3.26)$$

which must be solved for the weight matrix. Assuming for simplicity that all subchannels are active, i.e. $\Phi_f > 0$, the corresponding solution is given by [12]

$$\mathbf{W}_e^{1/2} = \mu^{1/2}(d\mathbf{D} + \mathbf{I})\mathbf{\Lambda}^{-1/2} \quad \text{with} \quad \mu^{1/2} = \frac{\text{tr}(\mathbf{\Lambda}^{-1/2}\mathbf{W}_e^{1/2})}{P_t + \text{tr}(\mathbf{\Lambda}^{-1})} \quad \text{and} \quad d = \frac{P_t}{\text{tr}(\mathbf{D}\mathbf{\Lambda}^{-1})}. \quad (3.27)$$

As a simple example, consider an application that requires all data streams to achieve the same reliability, using a fixed rate and an identical modulation and coding scheme for all streams. Within the present framework, this corresponds to imposing equal error rates and thus equal output SNRs on all eigen subchannels. The latter requirement can be satisfied trivially by selecting $\mathbf{D} = \mathbf{I}$. The resulting optimum precoder and combiner are then given by [12]

$$\mathbf{F} = \mathbf{V}\Phi_f \quad \text{with} \quad \Phi_f = \sqrt{\frac{P_t}{\text{tr}(\mathbf{\Lambda}^{-1})}} \mathbf{\Lambda}^{-1/2} \quad (3.28a)$$

$$\mathbf{G} = \Phi_g \mathbf{V}^H \mathbf{H}^H \mathbf{C}_n^{-1} \quad \text{with} \quad \Phi_g = \frac{\sqrt{P_t \text{tr}(\mathbf{\Lambda}^{-1})}}{P_t + \text{tr}(\mathbf{\Lambda}^{-1})} \mathbf{\Lambda}^{-1/2} \quad (3.28b)$$

This QoS design for equal error rates allocates more power to eigen subchannels with smaller channel gains and less power to stronger eigen subchannels, such that all active subchannels achieve the same output SNRs and MSEs. In contrast to the maximum-information-rate design, no eigen subchannels are dropped, regardless of the channel realization. Furthermore, it can be shown that the optimum precoder and combiner transform the transfer matrix of the system into a scaled identity matrix [12].

3.2.4 Unweighted MMSE Design

If the error weight matrix is chosen as $\mathbf{W}_e = \mathbf{I}$, the WMMSE problem reduces to the unweighted minimum mean squared error (MMSE) design. The closed-form solutions in (3.17a) and (3.17b) for the optimum precoder and combiner then minimize the sum of the expected squared symbol estimation errors across all eigen subchannels. Unlike the QoS design, this criterion does not enforce balanced performance across the individual subchannels, but improves the overall performance of the system. In particular, substituting $\mathbf{W}_e = \mathbf{I}$ into (3.24) shows that weaker subchannels generally exhibit larger MSEs, and therefore lower output SNRs, than stronger ones.

The corresponding power allocation follows from (3.18a) by setting $\mathbf{W}_e = \mathbf{I}$, which yields

$$\phi_{fi}^2 = \frac{1}{\sqrt{\lambda_i}} \left(\mu_w - \frac{1}{\sqrt{\lambda_i}} \right)_+ \quad \text{for} \quad i = 1, \dots, R_H \quad (3.29)$$

where ϕ_{fi}^2 is the i th element in the main diagonal of Φ_f and $\mu_w = \mu^{-1/2}$ is a water level chosen to satisfy the total power constraint with equality. This expression has exactly the same form as the weighted waterfilling solution derived in Appendix A. In particular, it is obtained from (A.6) by identifying the allocated power with $p_i = \phi_{fi}^2$, the channel weight with $\alpha_i = \lambda_i^{-1/2}$, and the channel-gain-to-noise ratio with $\gamma_i = \lambda_i$. Hence, the power allocation can be computed efficiently using Algorithm 1 with these parameter choices.

As in the maximum-information-rate design, eigen subchannels whose gains fall below a threshold receive no power. Among the remaining active subchannels, however, the weighting factor $\alpha_i = \lambda_i^{-1/2}$ gives relatively more emphasis to weaker subchannels than in the standard sum-rate case with $\alpha_i = 1$. This reflects the fact that the present design minimizes the total sum MSE, rather than maximizing the information rate.

3.3 Performance Evaluation

The preceding section derived optimal linear precoders and combiners for the SU-MIMO channel under several design criteria. This section evaluates the performance of the maximum information rate design through Monte Carlo simulations. As established in Section 3.2.2, this design is equivalent to the capacity-achieving SVD-based precoder and combiner with waterfilling power allocation.

All simulations are conducted under the following assumptions. The channel is modeled as an i.i.d. Rayleigh block fading channel, whose statistical model is derived in Appendix C. Each entry of the channel matrix $\mathbf{H}$ is drawn independently from $\mathcal{CN}(0, 1)$, and the channel is held constant within a transmission block and drawn independently across blocks. The reported performance metrics are averaged over at least 100 independent channel realizations. Perfect channel state information (CSI) is assumed at both the TX and the RX, so that the precoder and combiner are computed from the exact channel matrix.

The following subsections examine the effect of different system configurations and design parameters on the system performance. At the end of this section, analytical results are included to serve as a cross-check for the simulation curves. All results are presented as a function of the SNR, in terms of the total bit error rate (BER) and the total information bit rate (IBR) of the system.

3.3.1 Bit Allocation Strategies

As a first experiment, the BER is evaluated as a function of the SNR for a 2×2 SU-MIMO system using three modulation schemes: quadrature amplitude modulation (QAM), phase shift keying (PSK), and pulse amplitude modulation (PAM), each carrying the same number of bits per symbol. Figure 3.4 shows the results under a fixed bit allocation, where the same modulation order is applied on all eigen subchannels across the entire SNR range. As expected, the BER decreases monotonically with increasing SNR for all three constellations. The BER curve falls most steeply for QAM, followed by PSK and then PAM, confirming that QAM reaches any given target BER at a strictly lower SNR than the other two schemes at equal spectral efficiency. QAM is therefore adopted as the modulation scheme throughout this dissertation, unless otherwise stated. The number of bits per symbol is always chosen to be even, so that the QAM constellation forms a regular square grid.

As established in Section 3.2.2, the maximum information rate design is particularly relevant in adaptive modulation systems, where the number of bits per symbol is not fixed but adapted to the instantaneous capacity of each eigen subchannel. Specifically, the bit allocation on subchannel i

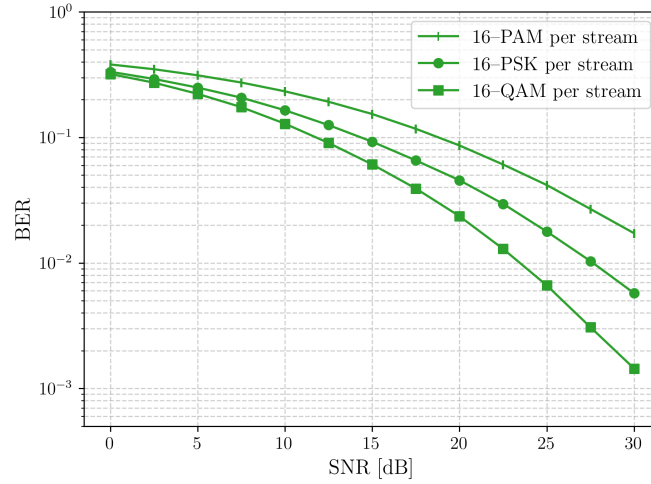


Figure 3.4: BER as a function of SNR for three SU-MIMO systems with 2 transmit and 2 receive antennas, each using a different modulation constellations on each stream: 16-PAM, 16-PSK, and 16-QAM.

is set to the largest even integer not exceeding its capacity. Figure 3.5 illustrates the resulting bit allocations for a 4×4 system at two different SNR values. At low SNR, the waterfilling solution concentrates almost all power on the strongest eigen subchannel, which is the only one capable of supporting a non-zero bit allocation. Because waterfilling maximizes the mutual information without accounting for the discrete nature of the bit allocation, a small amount of power is still directed to the weaker subchannels. Since their capacities fall below the 2-bit threshold, however, their bit allocations remain zero and the corresponding power is wasted. Jointly optimizing the power and bit allocation to avoid this inefficiency has been studied in [13, 14], but is not considered in this dissertation. At high SNR, more subchannels become active and support higher-order constellations, increasing the total IBR.

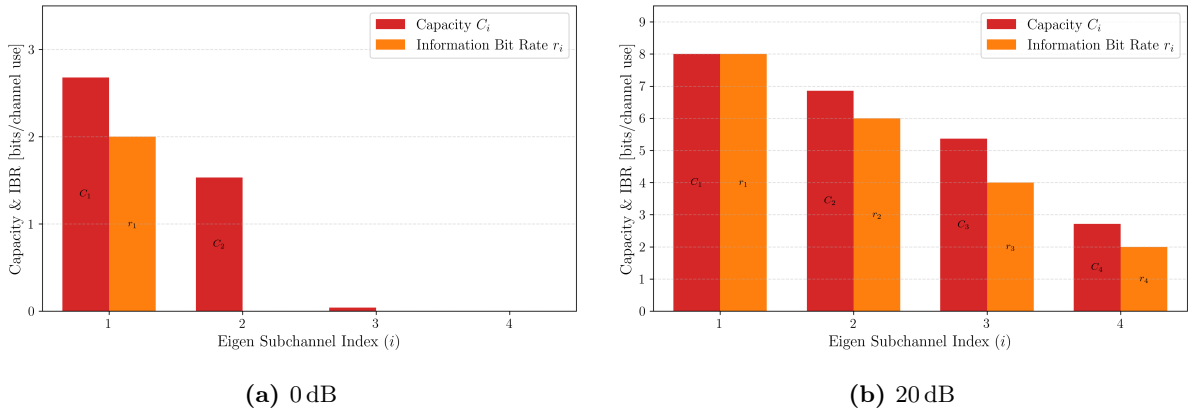


Figure 3.5: Adaptive bit allocation in different SNR regimes for a system with 4 transmit and 4 receive antennas. The number of bits per symbol on each stream is the largest even integer not exceeding the capacity of the corresponding eigen subchannel.

Figure 3.6 shows the BER and IBR as a function of SNR for this adaptive bit allocation strategy. In contrast to the fixed bit allocation case, the BER remains approximately constant across the entire SNR range, while the IBR increases steadily with SNR, as expected.

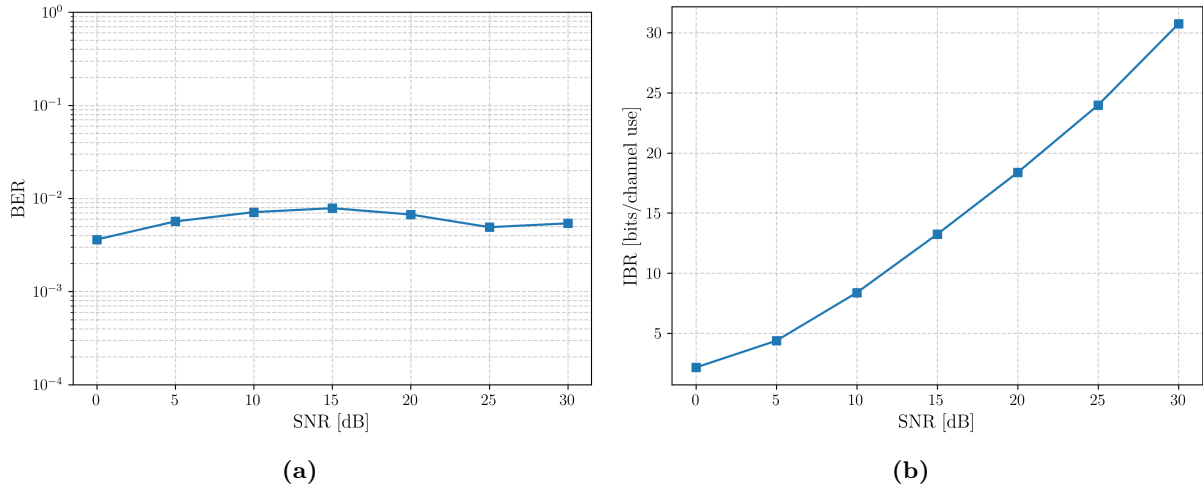


Figure 3.6: System performance with adaptive bit allocation for a system with 4 transmit and 4 receive antennas. The BER and IBR are shown as a function of SNR.

3.3.2 Impact of the Data Rate

The previous subsection assigned to each eigen subchannel the largest even number of bits not exceeding its full capacity, so that the system operated at approximately the maximum achievable rate. In practice, however, it may be preferable to transmit at a lower rate in exchange for a reduced BER. To quantify this tradeoff, we introduce a rate factor $r_f \in (0, 1]$ that specifies which fraction of the subchannel capacity is used for transmission. The bit allocation on subchannel i is then generalized to the largest even integer not exceeding $r_f C_i$, where C_i denotes the instantaneous capacity of that subchannel. The case $r_f = 1$ recovers the maximum-rate allocation of the previous subsection.

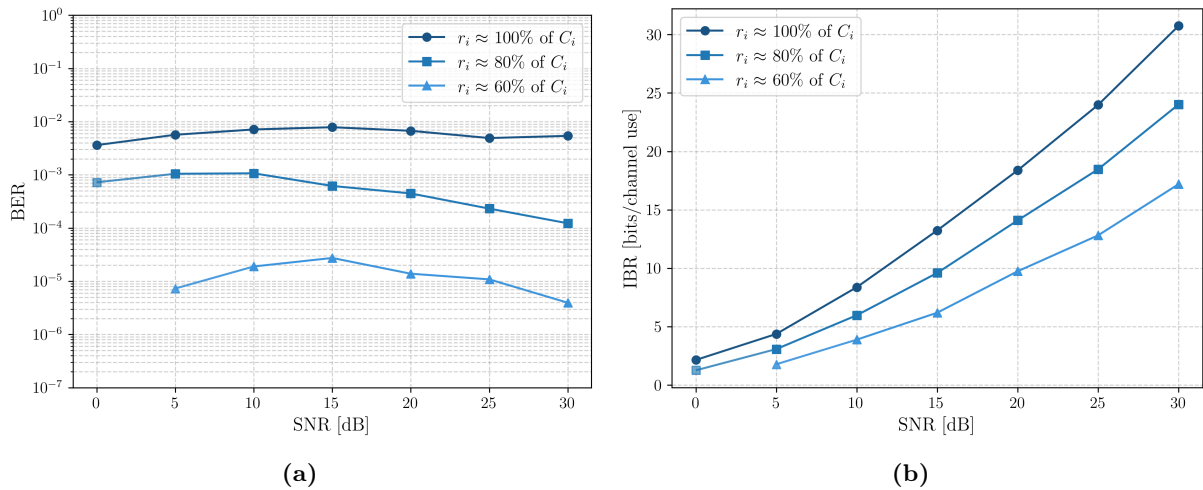


Figure 3.7: Illustration of the impact of the data rate on the system performance for a system with 4 transmit and 4 receive antennas using adaptive bit allocation. The (a) BER and (b) IBR are shown as a function of SNR, for three different rate factors.

Figure 3.7 shows the BER and IBR as a function of SNR for several values of r_f on a 4×4 system. As expected, reducing r_f lowers the IBR, because fewer bits are assigned to each subchannel. At the same time, the BER decreases, since the chosen modulation order moves further below the

subchannel capacity and the link operates with a larger margin. This illustrates an important tradeoff between throughput and reliability: operating below capacity sacrifices data rate in exchange for improved error performance.

For completeness, Figure 3.8 shows the analogous experiment under fixed bit allocation on a 2×2 system, where the same modulation order is applied on all eigen subchannels across the entire SNR range. The behavior is consistent: a smaller constellation size yields a lower BER at any given SNR. The slope of the BER curve is essentially unchanged by the constellation order.

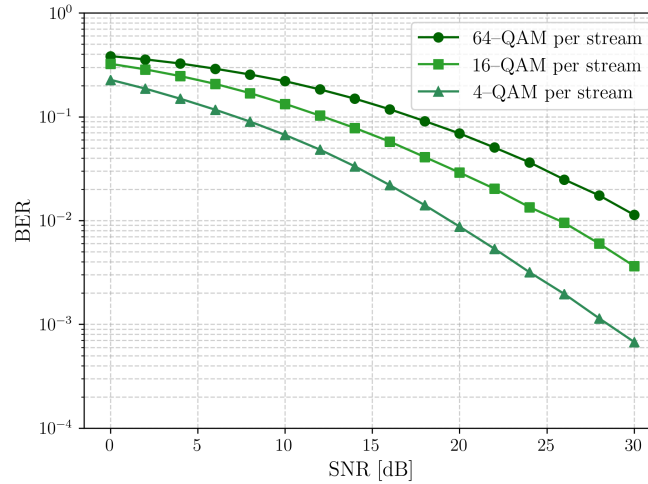


Figure 3.8: Illustration of the impact of the data rate on the system performance for a system with 2 transmit and 2 receive antennas using fixed bit allocation. The BER is shown as a function of SNR for three different modulation orders.

3.3.3 Impact of the Antenna Count

This subsection examines how the number of transmit and receive antennas affects system performance, keeping the rate factor fixed at $r_f = 1$. Figure 3.9 shows the BER and IBR as a function of SNR for several antenna configurations. The BER remains approximately constant across all configurations, consistent with the behavior observed in Section 3.3.1. By contrast, the maximum achievable rate, and thus the IBR, markedly with antenna count. Three distinct effects can be observed.

The first effect is a change in the slope of the IBR curve, which is determined by $\min(N_T, N_R)$. This follows directly from the channel rank: for an i.i.d. Rayleigh fading channel, $R_H \leq \min(N_T, N_R)$, so the number of parallel eigen subchannels cannot exceed the smaller antenna count. Consequently, adding antennas on both sides creates additional parallel streams and therefore increases the slope of the maximum achievable rate in function of the SNR.

The second effect is a vertical upward shift of the IBR curve when antennas are added only on the side with a larger antenna count while $\min(N_T, N_R)$ remains unchanged. This shift occurs because, for a fixed smaller dimension, the expected squared singular values of an $N_R \times N_T$ i.i.d. Gaussian matrix increase with the larger dimension [15]. The eigen subchannels thus experience better average channel conditions, yielding a higher IBR at the same SNR while leaving the slope unchanged.

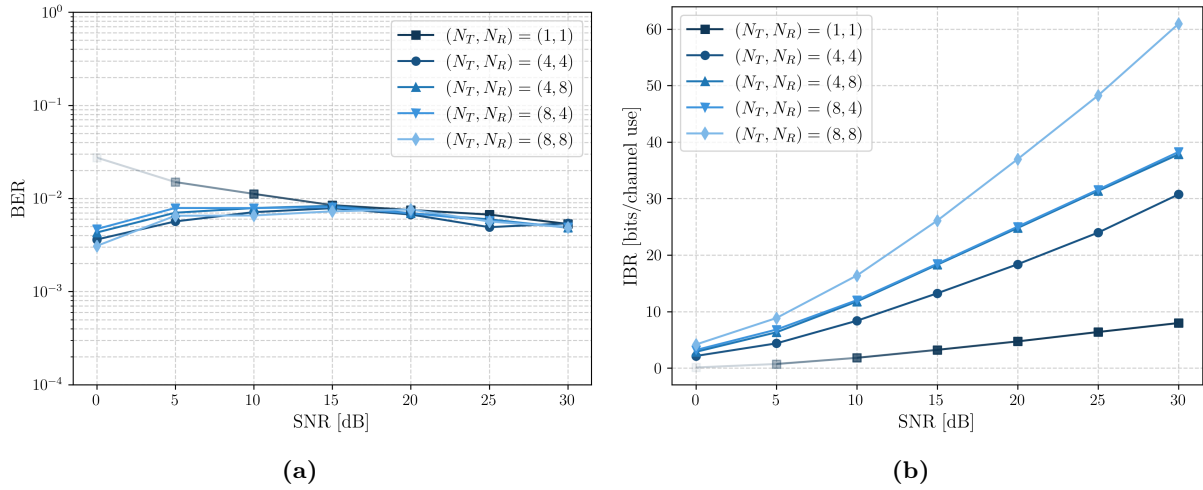


Figure 3.9: Illustration of the impact of the antenna count on the system performance for a system adaptive bit allocation. The (a) BER and (b) IBR are shown as a function of SNR, for five different antenna configurations.

Finally, configurations with the same $\min(N_T, N_R)$ and the same total number of antennas show no performance difference when the excess antennas are placed at the TX or at the RX. This symmetry follows since $\mathbf{H}\mathbf{H}^H$ and $\mathbf{H}^H\mathbf{H}$ have the same non-zero eigenvalues, so the singular-value distribution of an $N_R \times N_T$ i.i.d. Rayleigh channel is identical to that of its transpose.

As in the previous subsection, the rate–reliability tradeoff also applies here. Figure 3.10 illustrates this by adjusting the rate factor of each configuration so that all systems achieve approximately the same IBR at each SNR. Under this equal-rate comparison, systems with more antennas consistently achieve lower BER, because the additional spatial degrees of freedom support the same throughput with larger margins on the individual eigen subchannels.

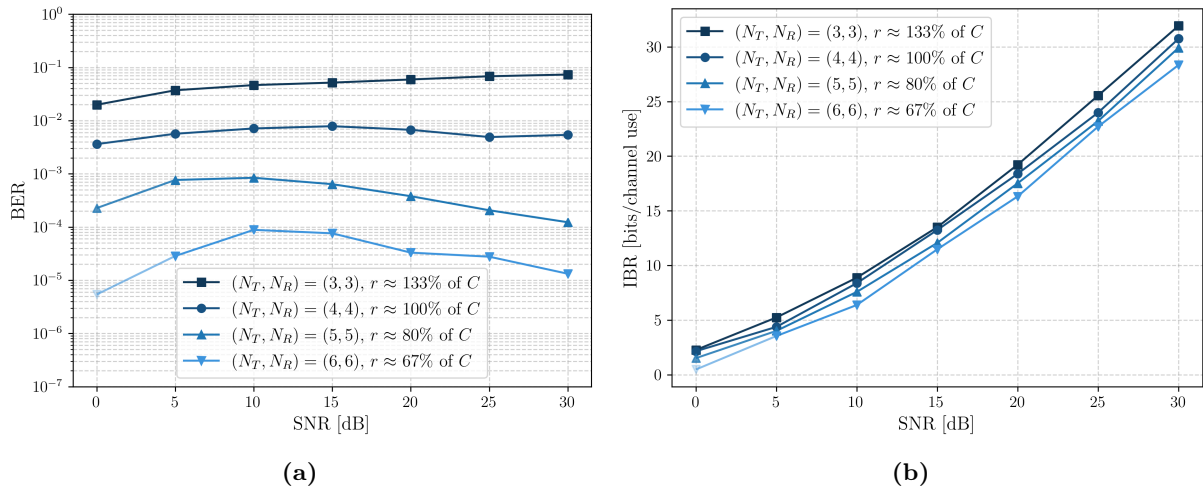


Figure 3.10: Illustration of the impact of the antenna count on the system performance for a system adaptive bit allocation. The (a) BER and (b) IBR are shown as a function of SNR, for five different antenna configurations. The rate factor of each configuration is adjusted so that all systems achieve approximately the same IBR at each SNR.

In summary, equipping both the TX and RX with more antennas enables either higher throughput or higher reliability, at the cost of increased hardware complexity and computational load.

3.3.4 Analytical Validation

As established in Section 3.1.2, the SVD-based precoder and combiner decouple the SU-MIMO channel into R_H parallel eigen subchannels, each of which can be treated as an independent SISO link. This decomposition enables an analytical evaluation of the total system BER.

Because the number of transmitted bits can differ across eigen subchannels, the total BER is computed as a weighted average of the subchannel BERs, with weights proportional to the corresponding IBRs:

$$\text{BER} = \frac{1}{\text{IBR}} \sum_{i=1}^{R_H} \text{IBR}_i \text{BER}_i. \quad (3.30)$$

The BER on each eigen subchannel equals the BER of a scalar additive white Gaussian noise (AWGN) channel. For subchannel i , the effective signal power is determined by the allocated transmit power scaled by the corresponding subchannel gain, while the noise power remains unchanged. The BER of a general scalar AWGN channel is derived in Appendix B under the union upper bound and the nearest-neighbour approximation.

Figure 3.11 compares the analytical expressions with the simulated BER curves for a 4×4 system under adaptive and fixed bit allocation. In both cases, the nearest-neighbour approximation and the union upper bound closely track the simulation results, which validates the simulation implementation.

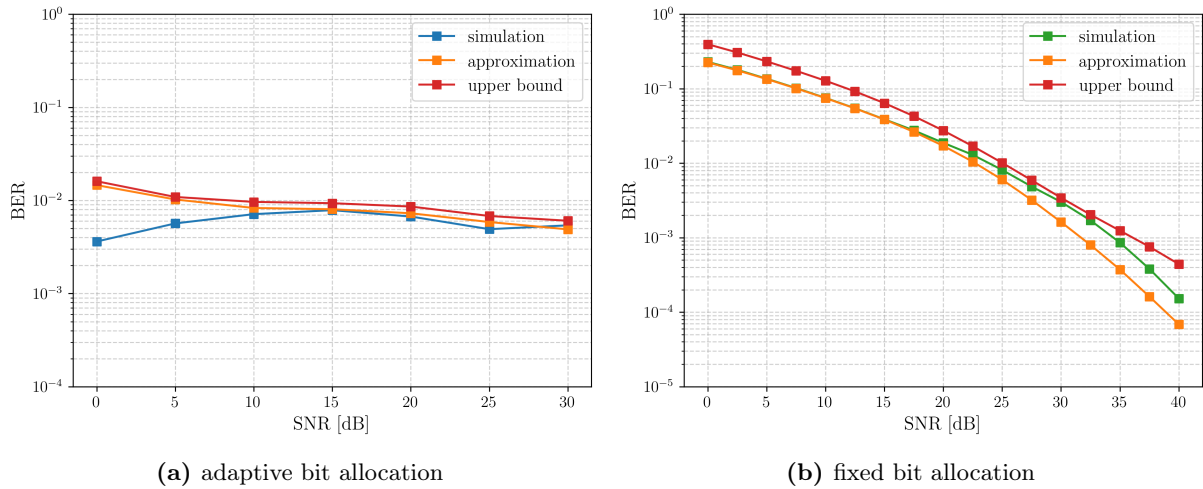


Figure 3.11: Analytical validation of the system performance simulation results for a system with 4 transmit and 4 receive antennas. The BER is shown as a function of SNR for two different bit allocation strategies: (a) adaptive bit allocation with a unit rate factor; and (b) fixed bit allocation, where all subchannels use 4-QAM across the entire SNR range. In both cases, the simulation results are compared with the union upper bound and nearest neighbour approximation derived in Appendix B.

Part II

Precoding for Satellite Communications

Part III

Appendices

Appendix A

Waterfilling for Power Allocation

In this appendix, we solve the problem of optimal power allocation across independent scalar Gaussian channels using the method of Lagrangian multipliers. The key principles of this method are explained alongside the derivation, because it is also employed to solve other optimization problems encountered in this dissertation. The solution to the optimal power allocation problem gives rise to the waterfilling algorithm, which is discussed at the end of this appendix.

A.1 Optimal Power Allocation

Consider N independent SISO Gaussian transmission channels. Optimal power allocation across these channels can be formulated as a constrained maximization problem. The objective is to distribute the total available transmit power among the different channels such that the combined achievable rate is maximized.

Consider the following general optimization problem:

$$\begin{aligned} \max_{\{p_n\}} \quad & \sum_{n=1}^N \alpha_n \log_2 (1 + \gamma_n p_n) \\ \text{s. t.} \quad & \sum_{n=1}^N p_n = P_t \quad \text{and} \quad p_n \geq 0 \quad \forall n = 1, \dots, N \end{aligned} \tag{A.1}$$

where p_n denotes the power allocated to channel n , and P_t denotes the total available transmit power. The coefficients α_n are non-negative weighting factors that allow different channels to be prioritized unequally, whereas $\gamma_n > 0$ denotes the ratio between the channel gain and the noise power of channel n .

To solve the constrained optimization problem in (A.1), the method of Lagrangian multipliers is employed. The core idea of this method is to convert a constrained optimization problem into an unconstrained one by absorbing the constraints into a modified objective function, known as the Lagrangian. This is achieved by introducing auxiliary variables, called Lagrange multipliers, one for each constraint, which penalize violations of the constraints.

Since the optimization problem in (A.1) involves both an equality constraint (total power budget) and inequality constraints (non-negativity of the individual power allocations), both types must be incorporated into the Lagrangian. Equality constraints enter with an unconstrained multiplier $\mu \in \mathbb{R}$, whereas inequality constraints of the form $p_n \geq 0$ are rewritten as $-p_n \leq 0$ and enter with a non-negative multiplier $\lambda_n \geq 0$, ensuring that the penalty is only active when the constraint is binding. The resulting Lagrangian is

$$\mathcal{L}(\mathbf{p}, \mu, \boldsymbol{\lambda}) = \sum_{n=1}^N \alpha_n \log_2(1 + \gamma_n p_n) - \mu \left(\sum_{n=1}^N p_n - P_t \right) + \sum_{n=1}^N \lambda_n p_n \quad (\text{A.2})$$

where μ is the Lagrange multiplier associated with the total power constraint, and λ_n are the multipliers associated with the non-negativity constraints on the individual power allocations.

Since the objective function is concave and all constraints are affine, the optimization problem is convex. For convex problems, the Karush-Kuhn-Tucker (KKT) conditions are both necessary and sufficient for a solution to be globally optimal [16, 17]. These conditions are given by

$$1. \text{ Stationarity:} \quad \nabla_{\mathbf{p}} \mathcal{L}(\mathbf{p}, \mu, \boldsymbol{\lambda}) = 0 \quad (\text{A.3a})$$

$$2. \text{ Primal feasibility:} \quad p_n \geq 0 \quad \text{and} \quad \sum_{n=1}^N p_n = P_t \quad (\text{A.3b})$$

$$3. \text{ Dual feasibility:} \quad \lambda_n \geq 0 \quad (\text{A.3c})$$

$$4. \text{ Complementary slackness:} \quad \begin{cases} p_n > 0 \implies \lambda_n = 0 \\ p_n = 0 \implies \lambda_n \geq 0 \end{cases} \quad (\text{A.3d})$$

From the KKT conditions, we can derive the optimal power allocation for each SISO channel. The derivation proceeds by successively applying the stationarity, complementary slackness, and primal feasibility conditions.

The stationarity condition requires that the gradient of the Lagrangian with respect to each optimization variable vanishes at the optimum. This expresses the balance between the marginal gain in the achievable rate and the penalties introduced by the power constraints. Applying this condition to the Lagrangian yields:

$$\nabla_{\mathbf{p}} \mathcal{L}(\mathbf{p}, \mu, \boldsymbol{\lambda}) = 0 \quad \Longleftrightarrow \quad \frac{\partial \mathcal{L}}{\partial p_n} = \frac{\alpha_n \gamma_n}{\ln(2)(1 + \gamma_n p_n)} - \mu + \lambda_n = 0, \quad \forall n = 1, \dots, N \quad (\text{A.4})$$

The complementary slackness condition links the non-negativity constraint on the allocated power to its associated Lagrange multiplier. When a positive amount of power is allocated to channel n , the corresponding multiplier λ_n is zero, and the stationarity condition reduces to:

$$\frac{\alpha_n \gamma_n}{\ln(2)(1 + \gamma_n p_n)} - \mu = 0 \quad (\text{A.5a})$$

$$\Longleftrightarrow p_n = \frac{\alpha_n}{\mu \ln(2)} - \frac{1}{\gamma_n} \quad (\text{A.5b})$$

The primal feasibility condition enforces the non-negativity of the power allocations. As a result, the above expression for p_n must be clipped to zero whenever it becomes negative:

$$p_n = \left(\frac{\alpha_n}{\mu \ln(2)} - \frac{1}{\gamma_n} \right)_+ \quad (\text{A.6})$$

Finally, the primal feasibility condition also requires the total allocated power to equal the available transmit power P_t . By substituting the expression for p_n from (A.6) into the total power constraint, we obtain an implicit equation for the Lagrangian multiplier μ :

$$\sum_{n=1}^N \left(\frac{\alpha_n}{\mu \ln(2)} - \frac{1}{\gamma_n} \right)_+ = P_t \quad (\text{A.7})$$

Since μ cannot be obtained in closed form, it is determined iteratively such that the total allocated power satisfies (A.7). Owing to the monotonic relationship between μ and the total allocated power, a bisection method is typically employed to compute the optimal value of μ numerically.

A.2 Waterfilling Algorithm

A more efficient alternative to the bisection method exploits the geometric structure of (A.7). Associate with each channel n a container of width α_n and floor height $1/(\alpha_n \gamma_n)$. The total available transmit power P_t is represented as water that is poured simultaneously into all containers. Because all containers share a common water level, channels with a lower floor and a wider container receive more water. A container remains empty if its floor height lies above the common water level, meaning no power is allocated to that channel. The water volume in each channel container is exactly equal to the power allocated to that channel under the optimal solution [18]. This interpretation is illustrated in Fig. A.1 for a system with $N = 4$ channels.

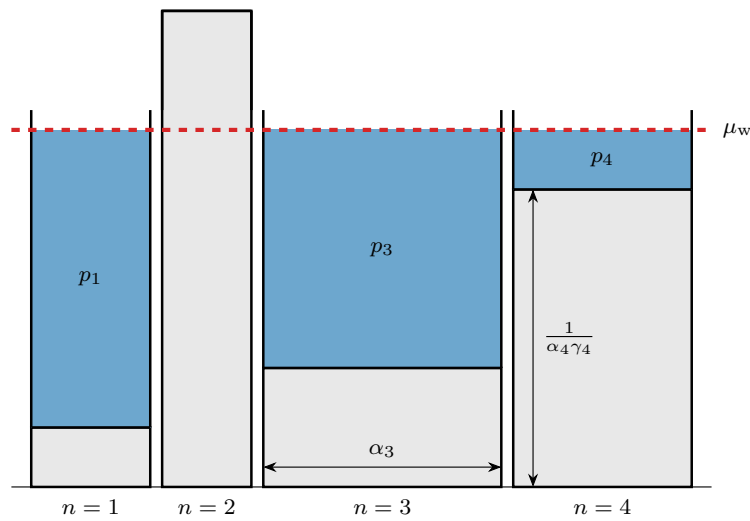


Figure A.1: A visualization of the waterfilling algorithm for optimal power allocation strategy in a system with $N = 4$ channels.

The complete procedure for determining the water level and the corresponding power allocation is summarized in Algorithm 1.

Algorithm 1 Waterfilling Algorithm for power allocation across Gaussian SISO subchannels

Require:

- The total transmit power: P_t ;
- The weight factor of each channel: α_n for $n = 1, \dots, N$;
- The ratios between channel gain and noise power: $\gamma_n = g_n^2/N_0$ for $n = 1, \dots, N$.

Ensure: The optimal power allocation p_n for $n = 1, \dots, N$.

Initialization: Determine the width and height of all channel containers.

Find the permutation $\mathcal{P} : n \rightarrow i$ between the original index n and the sorted index i that sorts the weighted ratios between channel gain and noise power $\alpha_n \gamma_n$ in decreasing order ($O(N \log N)$).

1: $w_i \leftarrow \alpha_{\mathcal{P}(n)}$, $h_i \leftarrow 1/(\alpha_{\mathcal{P}(n)} \gamma_{\mathcal{P}(n)})$

Iteration: Determine the optimal number of active channels ($O(\log N)$).

2: $N^{\text{LB}}, N^{\text{UB}} \leftarrow 0, N$

3: **while** $N^{\text{UB}} - N^{\text{LB}} > 1$ **do**

Update the number of active channels:

4: $N^{\text{iter}} \leftarrow \lfloor (N^{\text{UB}} + N^{\text{LB}})/2 \rfloor$

Compute the minimum power required to activate N^{iter} channels:

5: $P_{\min} \leftarrow \sum_{i=1}^{N^{\text{iter}}} w_i (h_{N^{\text{iter}}} - h_i)$

Update the bound on the number of active channels:

6: **if** $P_{\min} < P_t$ **then** $N^{\text{LB}} \leftarrow N^{\text{iter}}$ **else** $N^{\text{UB}} \leftarrow N^{\text{iter}}$

7: **end while**

Termination: Compute the power allocated to each channel.

Fill the N^{LB} active stream containers:

8: $p_i \leftarrow w_i (h_{N^{\text{LB}}} - h_i)$ for $i = 1, \dots, N^{\text{LB}}$

Distribute the remaining power among the active channels according to their weight:

9: $p_i \leftarrow p_i + (w_i / \sum_{i'=1}^{N^{\text{LB}}} w_{i'}) \cdot (P_t - \sum_{i'=1}^{N^{\text{LB}}} p_{i'})$ for $i = 1, \dots, N^{\text{LB}}$

Set the power of the inactive channels to zero:

10: $p_i \leftarrow 0$ for $i = N^{\text{LB}} + 1, \dots, N$

Use the inverse permutation $\mathcal{P}^{-1}$ to map the allocated power back to the original index n .

11: $p_n \leftarrow p_{\mathcal{P}^{-1}(i)}$

Return p_n for $n = 1, \dots, N$

A.3 Examples

Piecewise Frequency-Selective Channel

Consider a channel with a piecewise constant frequency response, partitioned into N non-overlapping subbands. Subband n spans the frequency interval $[B_{n-1}, B_n]$ and has a bandwidth $\beta_n = |B_n - B_{n-1}|$, a constant channel gain g_n , and experiences additive white Gaussian noise with power spectral density N_0 .

The capacity of this channel can then be written as

$$C = \max_{|S(f)|^2} \int_f \log_2 \left(1 + \frac{|H(f)|^2}{N_0} |S(f)|^2 \right) df \quad \text{s.t.} \quad \int_f |S(f)|^2 df \leq P_t \quad (\text{A.8a})$$

$$= \max_{|S(f)|^2} \sum_{n=1}^N \int_{B_{n-1}}^{B_n} \log_2 \left(1 + \frac{g_n}{N_0} |S(f)|^2 \right) df \quad \text{s.t.} \quad \int_f |S(f)|^2 df \leq P_t \quad (\text{A.8b})$$

$$= \max_{p_n} \sum_{n=1}^N \beta_n \log_2 \left(1 + \frac{g_n}{N_0} \frac{p_n}{\beta_n} \right) \quad \text{s.t.} \quad \sum_{n=1}^N p_n \leq P_t \quad (\text{A.8c})$$

where $S(f)$ denotes the power spectral density of the transmitted signal and $H(f)$ the frequency response of the channel. In the final step, Jensen's inequality and the concavity of the logarithm imply that the capacity-achieving power spectral density is constant within each subband. Accordingly, if p_n denotes the total power allocated to subband n , then $|S(f)|^2 = p_n/\beta_n$, for $f \in [B_{n-1}, B_n]$.

The optimization problem in (A.8c) is a special case of (A.1) with $\alpha_n = \beta_n$ and $\gamma_n = g_n/(N_0\beta_n)$. Hence, the capacity-achieving power allocation can be obtained by applying the waterfilling algorithm described in the previous section.

Single-User MIMO Channel

As shown in Section 3.1.1, a SU-MIMO channel can be decomposed into R_H independent parallel SISO eigen subchannels by means of the SVD of the channel matrix.

The capacity optimization problem takes the form

$$C = \max_{\{p_n\}} \sum_{n=1}^{N_s} \log_2 \left(1 + \frac{\sigma_n}{N_0} p_n \right) \quad \text{s.t.} \quad \sum_{n=1}^{N_s} p_n \leq P_t,$$

where p_n is the power allocated to the n -th eigen subchannel with channel gain σ_n and N_0 is the noise power on each receive antenna.

This is a special case of (A.1) with $\alpha_n = 1$ and $\gamma_n = \sigma_n/N_0$. Hence, the optimal power allocation, in the sense that it achieves the capacity, can be obtained by applying the waterfilling algorithm described in the previous section.

Appendix B

Analytical BER of an AWGN Channel

In this appendix, an analytical expression for the bit error rate (BER) of an uncoded transmission over an additive white Gaussian noise (AWGN) channel is derived.

The TX maps m_c independent and identically distributed (i.i.d.) bits onto a data symbol a drawn from the constellation $\mathcal{A}$, which contains $M = 2^{m_c}$ distinct symbols. The constellation is assumed to be normalized, i.e. $\mathbb{E}[|a|^2] = 1$. Let p denote the transmit power per symbol. The actual channel input is then given by $\sqrt{p}a$. Transmission takes place over an AWGN channel, so that the received signal can be written as

$$y = \sqrt{p}a + n, \quad (\text{B.1})$$

where n is a zero-mean CS complex Gaussian noise sample with variance N_0 . For convenience, the receiver works with the normalized decision variable $z = y/\sqrt{p}$, which allows the detection problem to be formulated directly in terms of the normalized constellation $\mathcal{A}$ and an equivalent noise sample w with variance N_0/p .

Based on the decision variable z and the constellation $\mathcal{A}$, the detector at the RX decides in favor of a symbol $\hat{a}$. The optimal detector is the one that minimizes the probability of decision error. For an AWGN channel and under the assumption of i.i.d. input bits, the optimal decision rule reduces to the minimum-distance decision rule, which selects the constellation symbol closest to z in Euclidean distance. Finally, the demapper maps the detected symbol $\hat{a}$ back to m_c bits, denoted by $\hat{\mathbf{b}}$. A block diagram of this transmission chain is shown in Figure B.1.

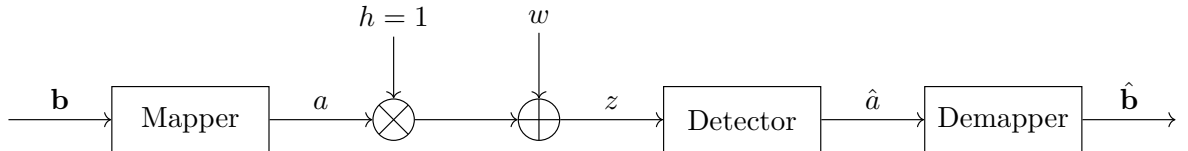


Figure B.1: Block diagram of an AWGN channel

The mapper, detector, and demapper are deterministic functions, whereas the transmitted symbol a and the noise sample n are random variables. Consequently, the decision variable z and the

detected symbol $\hat{a}$ are also random variables, both of which depend on the transmitted symbol and the noise realization. Conditioned on the transmitted symbol a , the decision variable is CS complex Gaussian with mean a and variance N_0/p , i.e.

$$z \mid a \sim \mathcal{CN}\left(a, \frac{N_0}{p}\right). \quad (\text{B.2})$$

The BER is defined as the probability that a transmitted bit is detected incorrectly. Equivalently, it can be expressed as the average number of bit errors per detected symbol, normalized by the number of bits per symbol:

$$\text{BER} = \frac{1}{\log_2(M)} \sum_{(\hat{\alpha}, \alpha) \in \mathcal{A}^2} N(\hat{\alpha}, \alpha) \text{Prob}[\hat{a} = \hat{\alpha} \wedge a = \alpha]. \quad (\text{B.3})$$

Here, $\mathcal{A}^2$ denotes the Cartesian product of the constellation with itself, and thus contains all possible transmitted-detected symbol pairs. The quantity $N(\hat{\alpha}, \alpha)$ denotes the number of bit errors incurred when the symbol α is transmitted and $\hat{\alpha}$ is detected. In other words, it equals the Hamming distance between the binary labels of α and $\hat{\alpha}$. Finally, $\text{Prob}[\hat{a} = \hat{\alpha} \wedge a = \alpha]$ is the joint probability that α is transmitted and $\hat{\alpha}$ is detected.

Since the transmission is uncoded and the input bits are i.i.d., all constellation symbols are equiprobable. The BER expression can therefore be simplified using Bayes' rule as

$$\text{BER} = \frac{1}{\log_2(M)} \sum_{(\hat{\alpha}, \alpha) \in \mathcal{A}^2} N(\hat{\alpha}, \alpha) \text{Prob}[\hat{a} = \hat{\alpha} \mid a = \alpha] \text{Prob}[a = \alpha] \quad (\text{B.4a})$$

$$= \frac{1}{M \log_2(M)} \sum_{(\hat{\alpha}, \alpha) \in \mathcal{A}^2} N(\hat{\alpha}, \alpha) \text{Prob}[\hat{a} = \hat{\alpha} \mid a = \alpha]. \quad (\text{B.4b})$$

B.1 Union Upper Bound

Consider the pairwise error event in which the transmitted symbol α is detected as a different symbol $\hat{\alpha}$, with $\hat{\alpha} \neq \alpha$. Under the minimum-distance decision rule, the detector decides in favor of $\hat{\alpha}$ if and only if the decision variable z falls inside the decision region associated with $\hat{\alpha}$. Hence,

$$\text{Prob}[\hat{a} = \hat{\alpha} \mid a = \alpha] = \text{Prob}[z \in \mathcal{D}(\hat{\alpha}) \mid a = \alpha], \quad (\text{B.5})$$

where $\mathcal{D}(\hat{\alpha})$ denotes the decision region of $\hat{\alpha}$, i.e. the set of all points in the complex plane that are closer to $\hat{\alpha}$ than to any other constellation symbol.

An exact evaluation of this conditional decision probability is generally cumbersome because the decision region may be bounded by several neighboring constellation points. A more tractable expression is obtained by replacing the full decision event by a necessary condition. Indeed, if $\hat{\alpha}$ is detected when α was transmitted, then the decision variable must at least be closer to $\hat{\alpha}$ than to α . This condition is necessary, but not sufficient, since yet another constellation point may

still be closer to z than $\hat{\alpha}$. Therefore,

$$\text{Prob}[\hat{a} = \hat{\alpha} \mid a = \alpha] \leq \text{Prob}[|z - \hat{\alpha}| < |z - \alpha| \mid a = \alpha]. \quad (\text{B.6})$$

Geometrically, this upper bound corresponds to the probability that the noise pushes the decision variable across the perpendicular bisector between α and $\hat{\alpha}$, rather than all the way into the decision region of $\hat{\alpha}$.

The right-hand side of (B.6) can be simplified as

$$\begin{aligned} & \text{Prob}[|z - \hat{\alpha}| < |z - \alpha| \mid a = \alpha] \\ &= \text{Prob}[|(\alpha + w) - \hat{\alpha}|^2 < |(\alpha + w) - \alpha|^2] \end{aligned} \quad (\text{B.7a})$$

$$= \text{Prob}[|w - (\hat{\alpha} - \alpha)|^2 - |w|^2 < 0] \quad (\text{B.7b})$$

$$= \text{Prob}[|\hat{\alpha} - \alpha|^2 - 2\Re\{w(\hat{\alpha} - \alpha)^*\} < 0] \quad (\text{B.7c})$$

$$= \text{Prob}\left[\Re\left\{w \frac{(\hat{\alpha} - \alpha)^*}{|\hat{\alpha} - \alpha|}\right\} > \frac{|\hat{\alpha} - \alpha|}{2}\right] \quad (\text{B.7d})$$

$$= Q\left(\sqrt{\frac{p}{2N_0}} |\hat{\alpha} - \alpha|\right) \quad (\text{B.7e})$$

In the last step, the conditional inequality is expressed as the tail probability of a real Gaussian random variable. Indeed, the quantity inside the probability operator is the projection of the CS complex Gaussian noise w onto a unit-magnitude complex direction. Such a multiplication only rotates w in the complex plane and therefore leaves its distribution unchanged. Consequently, the projected quantity is a zero-mean real Gaussian random variable with variance equal to half the variance of w , i.e. $N_0/(2p)$.

Substituting (B.7e) into (B.4) yields the following upper bound on the BER:

$$\text{BER} \leq \frac{1}{M \log_2(M)} \sum_{(\hat{\alpha}, \alpha) \in \mathcal{A}^2} N(\hat{\alpha}, \alpha) Q\left(\sqrt{\frac{p}{2N_0}} |\hat{\alpha} - \alpha|\right). \quad (\text{B.8})$$

This expression is known as the union upper bound.

B.2 Nearest Neighbour Approximation

The union upper bound in (B.8) sums over all possible transmitted-detected symbol pairs. For moderate to high SNR, however, the Q-function decays rapidly with increasing argument, so the dominant contributions arise from pairs $(\hat{\alpha}, \alpha)$ with the smallest inter-symbol distance. It is therefore common to retain only the nearest-neighbour terms in the sum, i.e. those located at the minimum Euclidean distance $d_{\min}$ between the constellation points in $\mathcal{A}$. Letting $\mathcal{S}(\alpha)$ denote the set of nearest neighbours of symbol α in $\mathcal{A}$, the approximation reads

$$\text{BER} \approx \frac{1}{M \log_2(M)} Q\left(\sqrt{\frac{p}{2N_0}} d_{\min}\right) \sum_{\alpha \in \mathcal{A}} \sum_{\hat{\alpha} \in \mathcal{S}(\alpha)} N(\hat{\alpha}, \alpha). \quad (\text{B.9})$$

For uncoded transmission, the optimal mapping from bits to symbol is given by Gray coding, which ensures that the binary labels of any two nearest-neighbour symbols differ in only one bit position, i.e. their Hamming distance is equal to one. Hence,

$$\text{BER} \approx \frac{1}{M \log_2(M)} Q\left(\sqrt{\frac{p}{2N_0}} d_{\min}\right) \sum_{\alpha \in \mathcal{A}} |\mathcal{S}(\alpha)|. \quad (\text{B.10})$$

where $|\mathcal{S}(\alpha)|$ denotes the number of nearest neighbours of symbol α . The remaining sum counts the total number of nearest-neighbour pairs in the constellation. Dividing by M gives the average number of nearest neighbours per symbol,

$$K = \frac{1}{M} \sum_{\alpha \in \mathcal{A}} |\mathcal{S}(\alpha)|, \quad (\text{B.11})$$

which is a constant that depends only on the type and size of the constellation. The values of K and $d_{\min}$ for all common modulation constellations, normalized to unit average symbol energy, are listed in Table B.1.

The final nearest-neighbour BER approximation reads

$$\text{BER} \approx \frac{K}{\log_2(M)} Q\left(\sqrt{\frac{p}{2N_0}} d_{\min}\right). \quad (\text{B.12})$$

Constellation	K	$d_{\min}$
M -PAM	$\frac{2(M-1)}{M}$	$\sqrt{\frac{12}{M^2-1}}$
M -PSK	$\begin{cases} 1, & M=2 \\ 2, & M>2 \end{cases}$	$2 \sin\left(\frac{\pi}{M}\right)$
M -QAM	$\frac{4(\sqrt{M}-1)}{\sqrt{M}}$	$\sqrt{\frac{6}{M-1}}$

Table B.1: Average number of nearest neighbours K and minimum inter-symbol distance $d_{\min}$ for common modulation constellations, normalized to unit average symbol energy.

Appendix C

Rayleigh Fading Channel Model

This appendix derives the Rayleigh channel model, which is commonly used to characterize wireless propagation in environments with many scatterers. Understanding the physical origin of this model provides useful intuition, since it serves as a fundamental building block for the more sophisticated channel models encountered throughout the dissertation. The derivation proceeds in three steps. First, the Doppler shift experienced by a moving RX receiving a signal reflected by a single scatterer is derived in Section C.1. Second, this result is extended to a large number of independent scatterers in Section C.2, and it is shown that the resulting channel gain converges to a circularly symmetric complex Gaussian random variable as the number of scatterers grows large. Finally, we explore the time-varying nature of the channel in Section ??, and derive the autocorrelation function and power spectral density of the fading process.

C.1 Doppler Shift

Consider a TX transmitting a baseband signal $x(t)$, modulated onto a carrier of frequency f_c . The signal reaches the RX after being reflected by a single scatterer, introducing an attenuation a_α . For now, the RX is assumed to be stationary, so the path length d_0 and hence the propagation delay $\tau_0 = d_0/c$ are constant. The received passband signal is given by

$$r_{\text{PB}}(t) = \Re\left\{a_\alpha x(t - \tau_0) e^{j2\pi f_c(t - \tau_0)}\right\}. \quad (\text{C.1})$$

Now suppose the RX moves with velocity v , and let α denote the angle between the direction of motion and the direction of the incoming wave. The component of velocity along the propagation path is $v \cos(\alpha)$, so the path length, and hence the propagation delay, become time-varying. This scenario is illustrated in Figure C.1.

The time-varying propagation delay is given by

$$\tau(t) = \frac{d_0 - v \cos(\alpha) t}{c} = \tau_0 - \frac{v \cos(\alpha)}{c} t. \quad (\text{C.2})$$

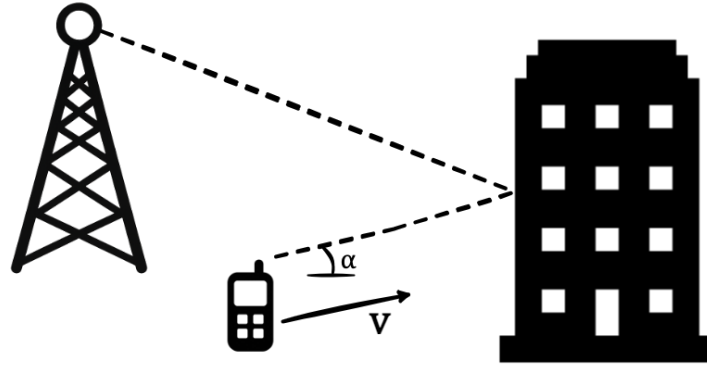


Figure C.1: An illustration of the scenario where the receiver experiences a Doppler shift due to its motion relative to the incoming signal reflected by a scatterer.

Substituting (C.2) into the expression for the received passband signal and expanding the phase term gives

$$r_{\text{PB}}(t) = \Re \left\{ a_{\alpha} x \left(\left(1 + \frac{v \cos(\alpha)}{c} \right) t - \tau_0 \right) e^{j2\pi(f_c + f_D \cos(\alpha))t} e^{-j2\pi f_c \tau_0} \right\}, \quad (\text{C.3})$$

where $f_D = v/\lambda$ is the maximum Doppler frequency and $\lambda = c/f_c$ is the carrier wavelength.

Two simplifying assumptions are now introduced to obtain a tractable baseband model. The first is the narrowband assumption: the bandwidth B of $x(t)$ is assumed to be much smaller than the carrier frequency, $B \ll f_c$. Since $x(t)$ varies slowly compared to the carrier, the small time dilation $v \cos(\alpha)/c \ll 1$ causes negligible distortion of the baseband envelope. The second assumption is perfect synchronization at the RX, so that the initial delay τ_0 can be neglected as well. Together, these two approximations give

$$x \left(\left(1 + \frac{v \cos(\alpha)}{c} \right) t - \tau_0 \right) \approx x(t). \quad (\text{C.4})$$

Applying both approximations and downconverting to baseband yields the received signal

$$y(t) = h(t) x(t), \quad \text{with} \quad h(t) = a_{\alpha} e^{j2\pi f_D \cos(\alpha) t + j\phi_{\alpha}}, \quad (\text{C.5})$$

where $\phi_{\alpha} = -2\pi f_c \tau_0$ is the initial carrier phase accumulated over the path length d_0 .

The channel $h(t)$ is a complex exponential with constant amplitude $|h(t)| = a_{\alpha}$ and a phase that advances linearly at the Doppler frequency $f_D \cos(\alpha)$. In practice, even sub-wavelength changes in d_0 cause variations of order 2π in ϕ_{α} , so it is customary to model ϕ_{α} as uniformly distributed over $[0, 2\pi)$ and independent across different scatterers.

C.2 CS Complex Gaussian Channel Gain

Consider the general scenario in which the signal from the TX reaches the RX via N independent scatterers. The scatterers are assumed to be uniformly distributed on a sphere around the RX,

so that the angles of arrival α_n are independent and identically distributed. Each scatterer n is further characterized by an initial phase ϕ_n , which is uniformly distributed over $[0, 2\pi)$ and independent across scatterers, as established in Section C.1. The individual attenuations are set to $a_n = 1/\sqrt{N}$, so that the total expected received power remains normalized to unity for any value of N . By superposition of the single-scatterer result in (C.5), the aggregate channel gain reads

$$h(t) = \sum_{n=1}^N a_n e^{j(2\pi f_D \cos(\alpha_n) t + \phi_n)}. \quad (\text{C.6})$$

To characterize the distribution of $h(t)$, the real and imaginary parts are analyzed separately. Let $\theta_n \triangleq 2\pi f_D \cos(\alpha_n) t$ denote the deterministic phase contribution of scatterer n at time t , and define

$$X(t) = \sum_{n=1}^N a_n \cos(\theta_n + \phi_n), \quad Y(t) = \sum_{n=1}^N a_n \sin(\theta_n + \phi_n), \quad (\text{C.7})$$

so that $h(t) = X(t) + jY(t)$. The mean, variance, and covariance of these components are computed below for a fixed time instant t .

Mean:

Since ϕ_n is uniformly distributed over $[0, 2\pi)$ and independent of α_n , the expectation of each summand can be evaluated conditionally on α_n using the law of total expectation:

$$\mathbb{E}[X(t)] = \sum_{n=1}^N \mathbb{E}[a_n \cos(\theta_n + \phi_n)] \quad (\text{C.8a})$$

$$= \sum_{n=1}^N \mathbb{E}[\mathbb{E}[a_n \cos(\theta_n + \phi_n) \mid \alpha_n]] \quad (\text{C.8b})$$

$$= \sum_{n=1}^N \mathbb{E}\left[a_n \int_0^{2\pi} \cos(\theta_n + \phi_n) \frac{1}{2\pi} d\phi_n\right] \quad (\text{C.8c})$$

$$= \sum_{n=1}^N \mathbb{E}[a_n \cdot 0] \quad (\text{C.8d})$$

Therefore, $\mathbb{E}[X(t)] = 0$. An identical argument with sine in place of cosine gives $\mathbb{E}[Y(t)] = 0$, and thus $\mathbb{E}[h(t)] = 0$.

Variance:

Since $\mathbb{E}[X(t)] = 0$, the variance reduces to the second moment, which can be expanded as

$$\text{Var}[X(t)] = \sum_{n=1}^N \sum_{m=1}^N \mathbb{E}[\mathbb{E}[a_n \cos(\theta_n + \phi_n) a_m \cos(\theta_m + \phi_m) \mid \alpha_n, \alpha_m]], \quad (\text{C.9})$$

where the law of total expectation is applied again to evaluate the expectation of each summand conditionally on α_n and α_m . For $n \neq m$, the independence of ϕ_n and ϕ_m allows the expectation

to factorize into a product of two zero-mean terms, so the cross terms vanish. For $n = m$,

$$\mathbb{E}[a_n^2 \cos^2(\theta_n + \phi_n) \mid \alpha_n] = a_n^2 \int_0^{2\pi} \cos^2(\theta_n + \phi_n) \frac{1}{2\pi} d\phi_n = \frac{1}{2} a_n^2. \quad (\text{C.10})$$

Summing the diagonal terms and substituting $a_n = 1/\sqrt{N}$ gives $\text{Var}[X(t)] = \frac{1}{2} \sum_{n=1}^N a_n^2 = 1/2$. An analogous calculation gives $\text{Var}[Y(t)] = 1/2$.

Covariance:

Since $\mathbb{E}[X(t)] = \mathbb{E}[Y(t)] = 0$, the covariance reduces to the cross moment, which can be expanded as

$$\text{Cov}[X(t), Y(t)] = \sum_{n=1}^N \sum_{m=1}^N \mathbb{E}[\mathbb{E}[a_n \cos(\theta_n + \phi_n) a_m \sin(\theta_m + \phi_m) \mid \alpha_n, \alpha_m]]. \quad (\text{C.11})$$

Similarly to the variance calculation, the cross terms vanish for $n \neq m$, and for $n = m$,

$$\mathbb{E}[a_n^2 \cos(\theta_n + \phi_n) \sin(\theta_n + \phi_n) \mid \alpha_n] = \frac{1}{2} a_n^2 \int_0^{2\pi} \sin(2\theta_n + 2\phi_n) \frac{1}{2\pi} d\phi_n = 0. \quad (\text{C.12})$$

Hence, $\text{Cov}[X(t), Y(t)] = 0$, confirming that the real and imaginary parts of the channel gain are uncorrelated.

Each summand in $X(t)$ is a bounded, zero-mean random variable, and the summands are mutually independent. By the Central Limit Theorem [19], as the number of scatterers grows large ($N \rightarrow \infty$), the distribution of $X(t)$ converges to a Gaussian with mean 0 and variance 1/2. The same holds for $Y(t)$. Since $X(t)$ and $Y(t)$ are independent Gaussian random variables, they are jointly Gaussian, and the channel gain $h(t) = X(t) + jY(t)$ is therefore a circularly symmetric complex Gaussian random variable with zero mean and unit variance:

$$h(t) \sim \mathcal{CN}(0, 1). \quad (\text{C.13})$$

A standard transformation to polar coordinates shows that the amplitude $|h(t)|$ follows a Rayleigh distribution, which is the origin of the model's name. The phase $\angle h(t)$ is uniformly distributed over $[0, 2\pi)$, as a direct consequence of the circular symmetry of $\mathcal{CN}(0, 1)$. Finally, the instantaneous power $|h(t)|^2$ follows an exponential distribution with unit mean.

C.3 Doppler Spectrum

Section C.2 characterized the distribution of $h(t)$ at a fixed time instant t . To describe the temporal variation of the fading process, the autocorrelation function (acf) and the power spectral density (PSD) of $h(t)$ are derived in this section.

As established in Section C.2, the channel gain at time t can be expressed as

$$h(t) = \sum_{n=1}^N a_n e^{j(2\pi f_D \cos(\alpha_n) t + \phi_n)}, \quad (\text{C.14})$$

where the expected power is normalized, i.e., $\sum_{n=1}^N \mathbb{E}[a_n^2] = 1$ and $\phi_n \sim \text{Uniform}(0, 2\pi)$ are independent across scatterers. The scatterers are further assumed to be uniformly distributed on a sphere around the RX, so that the angles of arrival are identically distributed as $\alpha_n \sim \text{Uniform}(0, 2\pi)$.

Autocorrelation Function

Let us calculate the acf $R_h(\tau) = \mathbb{E}[h(t)h^*(t - \tau)]$ of the fading process:

$$\begin{aligned}
R_h(\tau) &= \mathbb{E} \left[\left(\sum_{n=0}^N a_n \exp(j 2\pi f_D \cos(\alpha_n) t + j\phi_n) \right) \left(\sum_{m=0}^N a_m \exp(-j 2\pi f_D \cos(\alpha_m) (t - \tau) - j\phi_m) \right) \right] \\
&= \sum_{n=0}^N \sum_{m=0}^N \mathbb{E} [a_n \exp(j 2\pi f_D \cos(\alpha_n) t + j\phi_n) a_m \exp(-j 2\pi f_D \cos(\alpha_m) (t - \tau) - j\phi_m)] \\
&= \sum_{n=0}^N \sum_{m=0}^N \mathbb{E} [a_n a_m \exp(j 2\pi f_D (\cos \alpha_n - \cos \alpha_m) t) \exp(j 2\pi f_D \cos(\alpha_m) \tau) \exp(j\phi_n) \exp(-j\phi_m)] \\
&= \sum_{n=0}^N \sum_{m=0}^N \mathbb{E} [a_n a_m \exp(j 2\pi f_D (\cos \alpha_n - \cos \alpha_m) t) \exp(j 2\pi f_D \cos(\alpha_m) \tau)] \mathbb{E} [\exp(j(\phi_n - \phi_m))] \\
&\quad \bullet \mathbb{E} [\exp(j(\phi_n - \phi_m))] = \begin{cases} \mathbb{E} [e^{j(\phi_n - \phi_n)}] = 1 & \text{if } n = m \\ \mathbb{E} [e^{j\phi_n}] \mathbb{E} [e^{-j\phi_m}] = 0 & \text{if } n \neq m \end{cases} \\
&= \sum_{n=0}^N \mathbb{E} [a_n^2 e^{j 2\pi f_D \cos(\alpha_n) \tau}] \\
&= \sum_{n=0}^N \mathbb{E} [\mathbb{E} [a_n^2 e^{j 2\pi f_D \cos(\alpha_n) \tau} | \alpha_n]] \\
&= \sum_{n=0}^N \mathbb{E} \left[a_n^2 \int_0^{2\pi} e^{j 2\pi f_D \cos(\alpha_n) \tau} \frac{1}{2\pi} d\alpha_n \right] \\
&= \sum_{n=0}^N \mathbb{E} [a_n^2] \frac{1}{2\pi} \int_0^{2\pi} e^{j 2\pi f_D \cos(\alpha_n) \tau} d\alpha \\
&= \frac{1}{2\pi} \int_0^{2\pi} e^{j 2\pi f_D \cos(\alpha) \tau} d\alpha \\
&= J_0(2\pi f_D \tau).
\end{aligned}$$

The acf of the channel gain is therefore given by the zeroth-order Bessel function of the first kind, $J_0(2\pi f_D \tau)$, which quantifies the degree of correlation between the channel gain at two time instants separated by τ . The acf equals one at $\tau = 0$, since the channel gain is perfectly correlated with itself at zero lag, and decays as τ increases. The channel coherence time T_c is defined as the lag at which the acf drops to a prescribed threshold, here taken as $R_h(T_c) = 1/2$. For $\tau \ll T_c$, consecutive channel samples are highly correlated and the channel remains approximately constant over the interval. For $\tau \gg T_c$, the acf approaches zero and the samples are approximately uncorrelated. Figure C.2 illustrates the acf as a function of the normalized time lag τ/T_c .

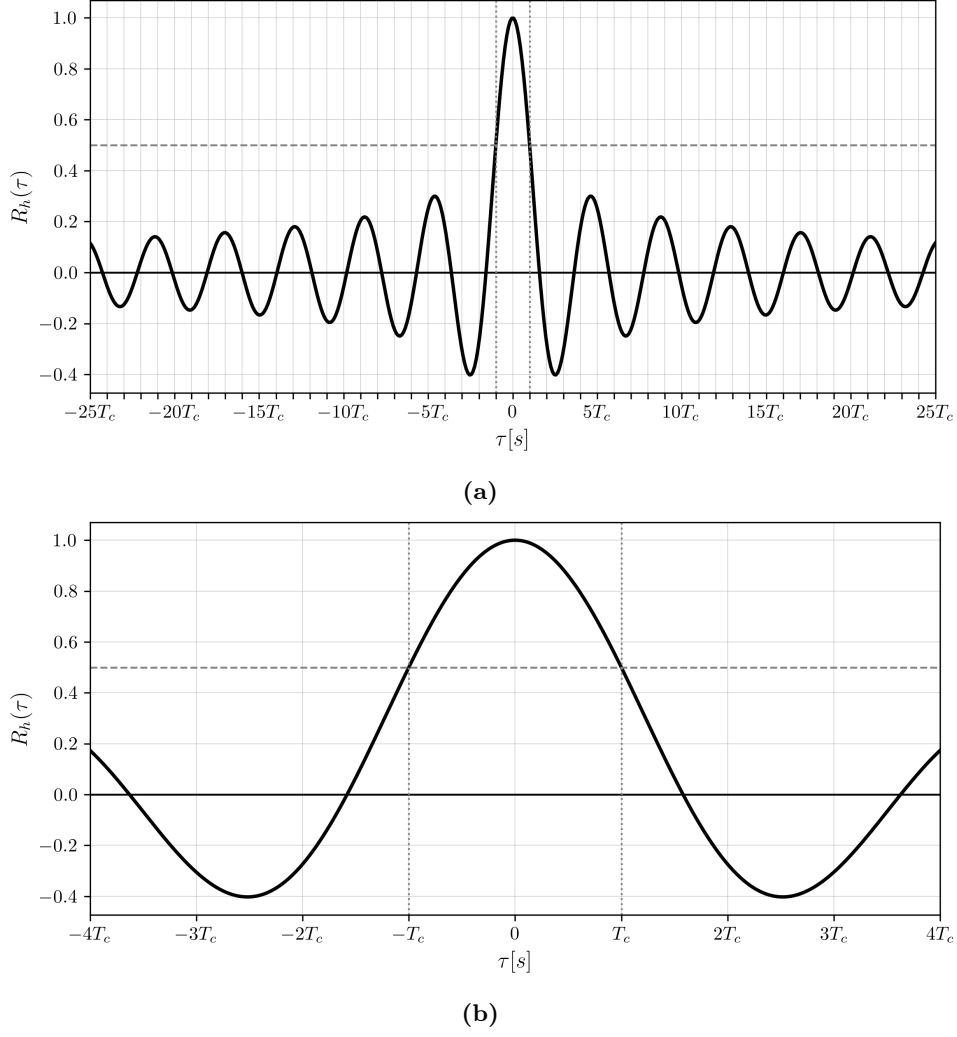


Figure C.2: The autocorrelation function of the channel gain fading process, relative to the coherence time T_c . (a) shows the autocorrelation for a maximum time lag of $25T_c$, while (b) zooms in on the region around the coherence time.

The relationship between T_c and f_D is obtained as follows:

$$R_h(T_c) = J_0(2\pi f_D T_c) = \frac{1}{2} \quad \Rightarrow \quad T_c \approx \frac{0.25}{f_D}. \quad (\text{C.15})$$

The coherence time is therefore inversely proportional to the maximum Doppler frequency. This makes intuitive sense: the faster the RX moves, and thus the higher the Doppler frequency $f_D = v/\lambda$, the more rapidly the channel fluctuates, and hence the shorter the coherence time.

When the symbol duration is much smaller than the coherence time ($T_s \ll T_c$), the channel gain is approximately constant over many symbols. In this slow fading regime, it is common to adopt the block fading model, in which $h(t)$ is assumed to be constant over a block of L symbols and drawn independently from $\mathcal{CN}(0, 1)$ for each successive block.

Power Spectral Density

By the Wiener-Khinchin theorem, the PSD of a wide-sense stationary process equals the Fourier transform of its acf. Let us compute the PSD $S_h(f) = \mathcal{F}\{R_h(\tau)\}$ of the fading process:

$$S_h(f)$$

$$\begin{aligned} &= \int_{-\infty}^{+\infty} R_h(\tau) e^{-j2\pi f\tau} d\tau \\ &= \int_{-\infty}^{+\infty} \left(\frac{1}{2\pi} \int_0^{2\pi} e^{j2\pi f_D \cos(\alpha)\tau} d\alpha \right) e^{-j2\pi f\tau} d\tau \\ &= \frac{1}{2\pi} \int_0^{2\pi} \left(\int_{-\infty}^{+\infty} e^{j2\pi f_D \cos(\alpha)\tau} e^{-j2\pi f\tau} d\tau \right) d\alpha \\ &= \frac{1}{2\pi} \int_0^{2\pi} \left(\int_{-\infty}^{+\infty} e^{-j2\pi (f - f_D \cos(\alpha))\tau} d\tau \right) d\alpha \end{aligned}$$

- definition Dirac delta function:

$$\begin{aligned} \delta(\xi) &= \int_{-\infty}^{+\infty} e^{-j2\pi \xi t} dt \iff \int_{-\infty}^{+\infty} e^{-j2\pi (f - f_D \cos(\alpha))\tau} d\tau = \delta(f - f_D \cos(\alpha)) \\ &= \frac{1}{2\pi} \int_0^\pi \delta(f - f_D \cos(\alpha)) d\alpha + \frac{1}{2\pi} \int_\pi^{2\pi} \delta(f - f_D \cos(\alpha)) d\alpha \end{aligned}$$

- substitution:

$$\begin{aligned} u &= f_D \cos(\alpha) \\ \Leftrightarrow du &= -f_D \sin(\alpha) d\alpha \\ \Leftrightarrow du &= -f_D \operatorname{sgn}(\sin \alpha) \sqrt{1 - \cos^2(\alpha)} d\alpha \\ \Leftrightarrow du &= -f_D \operatorname{sgn}(\sin \alpha) \sqrt{1 - \left(\frac{u}{f_D}\right)^2} d\alpha \\ \alpha = 0, \alpha = 2\pi &\implies u = f_D; \quad \alpha = \pi \implies u = -f_D \\ &= \frac{1}{2\pi} \int_{f_D}^{-f_D} \frac{\delta(f - u)}{-f_D \sqrt{1 - \left(\frac{u}{f_D}\right)^2}} du + \frac{1}{2\pi} \int_{-f_D}^{f_D} \frac{\delta(f - u)}{f_D \sqrt{1 - \left(\frac{u}{f_D}\right)^2}} du \\ &= \frac{1}{\pi} \int_{-f_D}^{f_D} \frac{\delta(f - u)}{f_D \sqrt{1 - \left(\frac{u}{f_D}\right)^2}} du \\ &= \frac{1}{\pi} \frac{1}{f_D \sqrt{1 - \left(\frac{u}{f_D}\right)^2}} \Big|_{u=f}, \quad \text{for } |f| < f_D \\ &= \frac{1}{\pi f_D \sqrt{1 - \left(\frac{f}{f_D}\right)^2}}, \quad \text{for } |f| < f_D \end{aligned}$$

The resulting PSD is known as the Doppler spectrum. It is nonzero only for $|f| \leq f_D$, confirming that the fading process is band-limited to the maximum Doppler frequency: since no scattering angle can produce a Doppler shift exceeding $\pm f_D$. The total Doppler spread equals $2f_D$.

The spectrum exhibits a characteristic “bathtub” shape, which is a direct consequence of the mapping $f = f_D \cos(\alpha)$. Near $\alpha = 0$ and $\alpha = \pi$, $\cos(\alpha)$ varies slowly as a function of α , so a

wide range of angles maps to a narrow frequency band near $\pm f_D$, concentrating power at the spectral edges. Near $\alpha = \pi/2$, $\cos(\alpha)$ varies rapidly, so comparatively few angles contribute per unit frequency around $f = 0$. Figure C.3 shows the resulting Doppler spectrum.

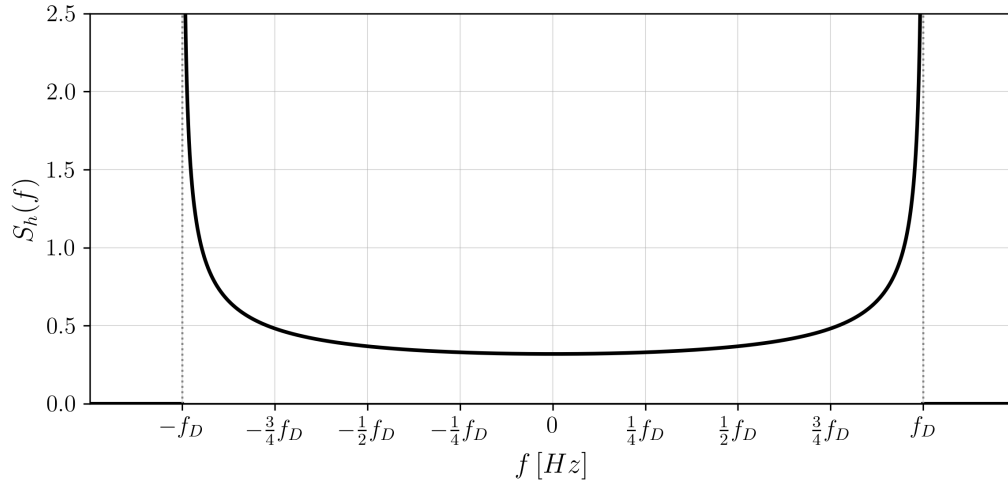


Figure C.3: The Doppler spectrum, relative to the maximum Doppler frequency f_D .

Appendix D

Computer Simulation of Time–Correlated Random Processes

When generating channel realizations for a time-varying channel, it is necessary to simulate a zero-mean complex Gaussian random process with a prescribed correlation in time. This appendix presents two practical methods for simulating time–correlated random processes on a computer. The first method uses the autocorrelation function as input and is based on the Cholesky decomposition of the covariance matrix. The second method uses the power spectral density as input and is based on filtering a white Gaussian process with a linear time-invariant system. In the simulation framework developed for this dissertation, both methods are employed to generate time–correlated channel realizations.

D.1 Cholesky Decomposition Method

The Cholesky decomposition method for generating a zero-mean unit-variance complex Gaussian process $\mathbf{h}$ of length N with a specified autocorrelation function $R_h(\tau)$ consists of the following steps:

Step 1: Build the $N \times N$ covariance matrix $\mathbf{C}$ with entries $C_{i,j} = R_h((i - j)T_s)$.

Step 2: Compute the Cholesky decomposition of the covariance matrix $\mathbf{C} = \mathbf{L}\mathbf{L}^H$, where $\mathbf{L}$ is a lower triangular matrix.

Step 3: Generate a column vector $\mathbf{w}$ of N i.i.d. white complex Gaussian random variables with zero mean and unit variance.

Step 4: Obtain the desired Gaussian process as $\mathbf{h} = \mathbf{L}\mathbf{w}$.

We show that the Cholesky decomposition method effectively generates a zero-mean unit-variance complex Gaussian process $\mathbf{h}$ of length N with a specified autocorrelation function $R_h(\tau)$.

First, let us verify that the generated process $\mathbf{h}$ is a complex Gaussian process with zero mean and the specified autocorrelation function.

Each element $h_n = \sum_{k=1}^n L_{k,n} w_k$ of the process $\mathbf{h}$ is a complex Gaussian random variable, since w_k are i.i.d. complex Gaussian random variables and the linear combination of complex Gaussian random variables is itself also a complex Gaussian random variable. Let us now calculate the mean and covariance matrix of the generated process $\mathbf{h}$.

The expected value of each element h_n is given by:

$$\mathbb{E}[h_n] = \mathbb{E} \left[\sum_{k=1}^n L_{k,n} w_k \right] = \sum_{k=1}^n L_{k,n} \mathbb{E}[w_k] = 0 \quad (\text{D.1})$$

The covariance matrix of $\mathbf{h}$ is given by:

$$\begin{aligned} \text{Cov}[\mathbf{h}] &= \mathbb{E}[(\mathbf{h} - \mathbb{E}[\mathbf{h}])(\mathbf{h} - \mathbb{E}[\mathbf{h}])^H] = \mathbb{E}[\mathbf{h}\mathbf{h}^H] \\ &= \mathbb{E}[(\mathbf{L}\mathbf{w})(\mathbf{L}\mathbf{w})^H] = \mathbb{E}[\mathbf{L}\mathbf{w}\mathbf{w}^H\mathbf{L}^H] = \mathbf{L} \mathbb{E}[\mathbf{w}\mathbf{w}^H] \mathbf{L}^H \\ &= \mathbf{L} \mathbf{I}_N \mathbf{L}^H = \mathbf{L}\mathbf{L}^H = \mathbf{C} \end{aligned} \quad (\text{D.2})$$

From (D.1) and (D.2), the mean of the generated process $\mathbf{h}$ is zero and the covariance of $\mathbf{h}$ matches the specified autocorrelation function $R_h(\tau)$ by construction. This confirms that the Cholesky decomposition method effectively generates a zero-mean unit-variance complex Gaussian process with the desired correlation properties:

$$\mathbf{h} \sim \mathcal{CN}(\mathbf{0}, \mathbf{C}) \quad (\text{D.3})$$

Second, let us demonstrate that the Cholesky decomposition of the covariance matrix $\mathbf{C}$ always exists. It is well-known that the Cholesky decomposition of a matrix $\mathbf{C}$ exists and is unique if and only if $\mathbf{C}$ is an Hermitian matrix and a positive-definite matrix, i.e.:

$$\exists! \mathbf{L} : \mathbf{C} = \mathbf{L}\mathbf{L}^H \iff \mathbf{C} = \mathbf{C}^H \quad \text{and} \quad \forall \mathbf{x} \in \mathbb{C}^N \setminus \{\mathbf{0}\} : \mathbf{x}^H \mathbf{C} \mathbf{x} > 0 \quad (\text{D.4})$$

Consider a autocorrelation function $R_h(\tau)$ of a wide-sense stationary process, i.e. R_h does not depend on the absolute time t , but only on the time difference τ . Therefore, the covariance matrix $\mathbf{C}$ is a real and symmetric Toeplitz matrix by construction. Thus, $\mathbf{C}$ is a Hermitian matrix.

We prove that $\mathbf{C}$ is positive-definite by contradiction. First, let us simplify the expression $\mathbf{x}^H \mathbf{C} \mathbf{x}$:

$$\begin{aligned} \mathbf{x}^H \mathbf{C} \mathbf{x} &= \sum_{i=1}^N \sum_{j=1}^N x_i^* C_{i,j} x_j = \sum_{i=1}^N \sum_{j=1}^N x_i^* R_h((i-j)T_s) x_j \\ &= \sum_{i=1}^N \sum_{j=1}^N x_i^* \left(\int_{-\infty}^{\infty} S_h(f) e^{j2\pi f(i-j)T_s} df \right) x_j \\ &= \int_{-\infty}^{\infty} S_h(f) \left(\sum_{i=1}^N x_i^* e^{j2\pi f iT_s} \right) \left(\sum_{j=1}^N x_j e^{-j2\pi f jT_s} \right) df \end{aligned}$$

$$\begin{aligned}
&= \int_{-\infty}^{\infty} S_h(f) \left| \sum_{i=1}^N x_i e^{-j2\pi f i T_s} \right|^2 df \\
&= \int_{-\infty}^{\infty} S_h(f) |X(f)|^2 df
\end{aligned} \tag{D.5}$$

We consider a autocorrelation function $R_h(\tau)$ that corresponds to a power spectral density that makes physical sense, i.e. $S_h(f)$ is positive for all frequencies f and strictly positive for a set of positive measure. Then, it is easy to see that $\mathbf{C}$ is positive semi-definite, since the integrand is non-negative for all frequencies f .

Assume that $\mathbf{C}$ is not positive-definite, then there exists a non-zero vector $\mathbf{x} \in \mathbb{C}^N$ such that $\mathbf{x}^H \mathbf{C} \mathbf{x} = 0$. For that to be true, the integrand must be zero, which implies that $X(f) = 0$ for all frequencies f where $S_h(f) > 0$, which is a set of positive measure. $X(f)$ can be seen as the $\mathcal{Z}$ -transform of $\mathbf{x}$ evaluated on the unit circle, which corresponds to a complex polynomial of degree at most $N - 1$. By the Fundamental Theorem of Algebra, such a nonzero polynomial has at most $N - 1$ roots. Therefore, $\mathbf{x}$ must be the zero vector, which contradicts the assumption. Hence, $\mathbf{C}$ is positive-definite.

D.2 Linear Filtering Method

The linear filtering method for generating a zero-mean unit-variance complex Gaussian process with a specified power spectral density $S_h(f)$ consists of the following steps:

Step 1: Evaluate the target PSD $S_h(f)$ on an FFT frequency grid of N_g uniformly spaced normalized frequencies $\nu_k \in [-1/2, 1/2)$. To avoid aliasing, the sampling period T_s must satisfy the Nyquist condition $f_D T_s < 1/2$, where f_D denotes the maximum Doppler frequency.

Step 2: Construct the frequency-domain shaping response by taking the nonnegative square root of the sampled PSD: $G[\nu_k] = \sqrt{S_h[\nu_k]}$.

Step 3: Apply an inverse discrete Fourier transform to the shaping response $G[\nu_k]$ in order to obtain the impulse response of a finite-length FIR filter, $g[n] = \mathcal{F}^{-1}\{G[\nu_k]\}$, and normalize this filter to unit energy.

Step 4: Generate a sequence of white proper complex Gaussian random variables $w[n]$ with zero mean and unit variance per sample.

Step 5: Obtain the desired time-correlated output process by convolving the white-noise sequence with the shaping filter: $h[n] = \sum_m g[m] w[n - m]$.

The underlying idea behind this method is that passing white complex Gaussian noise through a linear time-invariant filter with frequency response $G(f)$ yields an output process whose power spectral density is equal to $|G(f)|^2$. Consequently, by choosing $G(f) = \sqrt{S_h(f)}$, the output process is shaped so as to have the desired power spectral density. The derivation below makes this argument precise.

First, note that the output process $h(t)$ is a linear transformation of the white Gaussian input process $w(t)$ and is therefore itself a complex Gaussian process. Moreover, because the input process has zero mean and the filter is linear, the mean of the output process is also zero.

Let us now determine the autocorrelation function of the output process $h(t)$:

$$\begin{aligned}
R_h(\tau) &= \mathbb{E}[h(t) h^*(t - \tau)] \\
&= \mathbb{E} \left[\int_{-\infty}^{+\infty} g(u) w(t - u) du \int_{-\infty}^{+\infty} g(v) w^*(t - \tau - v) dv \right] \\
&= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(u) g^*(v) \mathbb{E}[w(t - u) w^*(t - \tau - v)] du dv \\
&\quad \bullet \quad \mathbb{E}[w(t - u) w^*(t - \tau - v)] \\
&\quad = \mathbb{E}[w(t - u) w^*((t - u) - (\tau - v - u))] = R_w(\tau + v - u) = \delta(\tau + v - u) \\
&= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(u) g^*(v) \delta(\tau + v - u) du dv \\
&= \int_{-\infty}^{+\infty} g(u) g^*(u - \tau) du \\
&= R_g(\tau)
\end{aligned} \tag{D.6}$$

Equation (D.6) shows that the autocorrelation function of the output process is equal to the autocorrelation function of the shaping filter. In particular, because the filter is normalized to unit energy, the variance of the output process is given by $R_h(0) = R_g(0) = \int_{-\infty}^{+\infty} |g(u)|^2 du = 1$. Hence, the generated process has unit variance.

Next, we compute the Fourier transform of the autocorrelation function in order to obtain the power spectral density of the output process:

$$\begin{aligned}
S_h(f) &= \int_{-\infty}^{+\infty} R_g(\tau) e^{-j2\pi f\tau} d\tau \\
&= \int_{-\infty}^{+\infty} \left(\int_{-\infty}^{+\infty} g(u) g^*(u - \tau) du \right) e^{-j2\pi f\tau} d\tau \\
&= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(u) g^*(u - \tau) e^{-j2\pi f\tau} du d\tau \\
&= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(u) g^*(v) e^{-j2\pi f(u-v)} du dv \\
&= \left(\int_{-\infty}^{+\infty} g(u) e^{-j2\pi fu} du \right) \left(\int_{-\infty}^{+\infty} g^*(v) e^{j2\pi fv} dv \right) \\
&= G(f) G^*(f) \\
&= |G(f)|^2
\end{aligned} \tag{D.7}$$

Equation (D.7) confirms that the power spectral density of the output process equals the squared magnitude of the filter's frequency response. Therefore, choosing the shaping filter such that $G(f) = \sqrt{S_h(f)}$ ensures that the generated process has the desired power spectral density.

In a practical computer implementation, the process cannot be generated in continuous time and with an infinite-length filter. Instead, the shaping filter is approximated by a finite-length discrete-time filter obtained from sampled values of the target PSD via the inverse discrete Fourier transform (DFT). It is important to choose the filter length N_g sufficiently large, so that truncation effects become negligible and the finite-length filter accurately approximates the target correlation properties of the process.

Appendix E

Calculations

In this appendix, we provide the calculations that are relevant to the main text but were omitted for brevity. The appendix is organized into separate sections that correspond to specific intermediate results used throughout the dissertation.

Total Transmit Power Constraint

The following calculations derive the total transmit power as a function of the precoding matrix for both the SU-MIMO and MU-MIMO system models. In both cases, the power allocation is assumed to be incorporated into the precoding matrix, and the data symbols are assumed to be zero-mean, mutually uncorrelated, and normalized, i.e. $\mathbb{E}[\mathbf{a}\mathbf{a}^H] = \mathbf{I}_{N_S}$.

Consider the SU-MIMO system model as described in Section 2.2.1. The total transmit power is constraint is given by:

$$\mathbb{E}[\|\mathbf{x}\|^2] = \text{tr}(\mathbf{F}^H \mathbf{F}) \leq P_t.$$

Calculation:

$$\begin{aligned} P_t &\geq \mathbb{E}[\|\mathbf{x}\|^2] = \mathbb{E}[\mathbf{x}^H \mathbf{x}] = \mathbb{E}[(\mathbf{F}\mathbf{a})^H (\mathbf{F}\mathbf{a})] = \mathbb{E}[\mathbf{a}^H \mathbf{F}^H \mathbf{F} \mathbf{a}] \\ &= \mathbb{E}[\text{tr}(\mathbf{a}^H \mathbf{F}^H \mathbf{F} \mathbf{a})] = \mathbb{E}[\text{tr}(\mathbf{F}^H \mathbf{F} \mathbf{a} \mathbf{a}^H)] \\ &= \text{tr}(\mathbb{E}[\mathbf{F}^H \mathbf{F} \mathbf{a} \mathbf{a}^H]) = \text{tr}(\mathbf{F}^H \mathbf{F} \mathbb{E}[\mathbf{a} \mathbf{a}^H]) \\ &= \text{tr}(\mathbf{F}^H \mathbf{F} \mathbf{I}_{N_S}) = \text{tr}(\mathbf{F}^H \mathbf{F}) \end{aligned}$$

Consider the MU-MIMO system model as described in Section 2.2.2. The total transmit power is constraint is given by:

$$\mathbb{E}[\|\mathbf{x}\|^2] = \sum_{k=1}^K \text{tr}(\mathbf{F}_k^H \mathbf{F}_k) \leq P_t.$$

Calculation:

$$\begin{aligned}
P_t &\geq \mathbb{E}[\|\mathbf{x}\|^2] = \mathbb{E}[\mathbf{x}^H \mathbf{x}] = \mathbb{E}\left[\left(\sum_{k=1}^K \mathbf{F}_k \mathbf{a}_k\right)^H \left(\sum_{k'=1}^K \mathbf{F}_{k'} \mathbf{a}_{k'}\right)\right] \\
&= \mathbb{E}\left[\sum_{k=1}^K \sum_{k'=1}^K (\mathbf{F}_k \mathbf{a}_k)^H \mathbf{F}_{k'} \mathbf{a}_{k'}\right] = \mathbb{E}\left[\sum_{k=1}^K \sum_{k'=1}^K \mathbf{a}_k^H \mathbf{F}_k^H \mathbf{F}_{k'} \mathbf{a}_{k'}\right] \\
&= \mathbb{E}\left[\sum_{k=1}^K \sum_{k'=1}^K \text{Tr}(\mathbf{a}_k^H \mathbf{F}_k^H \mathbf{F}_{k'} \mathbf{a}_{k'})\right] = \mathbb{E}\left[\sum_{k=1}^K \sum_{k'=1}^K \text{Tr}(\mathbf{F}_k^H \mathbf{F}_{k'} \mathbf{a}_{k'} \mathbf{a}_k^H)\right] \\
&= \sum_{k=1}^K \sum_{k'=1}^K \text{Tr}(\mathbf{F}_k^H \mathbf{F}_{k'} \mathbb{E}[\mathbf{a}_{k'} \mathbf{a}_k^H]) \\
&= \sum_{k=1}^K \sum_{k'=1}^K \text{Tr}(\mathbf{F}_k^H \mathbf{F}_{k'} \mathbf{I}_{N_S} \delta_{kk'}) \\
&= \sum_{k=1}^K \text{Tr}(\mathbf{F}_k^H \mathbf{F}_k)
\end{aligned}$$

Equivalent SU-MIMO Channel Representation

Applying the singular value decomposition of the channel matrix together with a change of variables yields an equivalent representation of the SU-MIMO channel. The following calculation shows how the original channel input-output relation is transformed into a set of parallel scalar Gaussian subchannels.

Calculation:

$$\begin{aligned}
\mathbf{y} &= \mathbf{H}\mathbf{x} + \mathbf{n} \\
\Leftrightarrow \mathbf{y} &= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H \mathbf{x} + \mathbf{n} \\
\Leftrightarrow \mathbf{U}\tilde{\mathbf{y}} &= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H \mathbf{V}\tilde{\mathbf{x}} + \mathbf{n} \\
\Leftrightarrow \tilde{\mathbf{y}} &= \mathbf{U}^H \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H \mathbf{V}\tilde{\mathbf{x}} + \mathbf{U}^H \mathbf{n} \\
\Leftrightarrow \tilde{\mathbf{y}} &= \mathbf{\Sigma}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}.
\end{aligned}$$

Mutual Information under Optimal Precoder and Combiner

The following calculation verifies that the MI between the transmitted data symbols $\mathbf{a}$ and the decision variables $\mathbf{z}$ equals the MI between the channel input $\mathbf{x}$ and the channel output $\mathbf{y}$ when the optimal precoder and combiner from (3.8) are employed. The result is obtained by substituting $\mathbf{F} = \mathbf{V}_{N_s} \sqrt{\mathbf{P}}$ and $\mathbf{G} = \mathbf{U}_{N_s}^H$ into (3.7b) and replacing the channel matrix by its SVD (3.1).

Calculation:

$$\begin{aligned}
I(\mathbf{a}; \mathbf{z}) &= \log_2 \left(\det \left(\mathbf{I}_{N_s} + \frac{1}{N_0} (\mathbf{U}_{N_s}^H \mathbf{U}_{N_s})^{-1} (\mathbf{U}_{N_s}^H (\mathbf{U} \mathbf{\Sigma} \mathbf{V}^H) \mathbf{V}_{N_s} \sqrt{\mathbf{P}}) (\mathbf{U}_{N_s}^H (\mathbf{U} \mathbf{\Sigma} \mathbf{V}^H) \mathbf{V}_{N_s} \sqrt{\mathbf{P}})^H \right) \right) \\
&= \log_2 \left(\det \left(\mathbf{I}_{N_s} + \frac{1}{N_0} \mathbf{I}_{N_s} (\mathbf{U}_{N_s}^H \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H \mathbf{V}_{N_s} \sqrt{\mathbf{P}}) (\mathbf{U}_{N_s}^H \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H \mathbf{V}_{N_s} \sqrt{\mathbf{P}})^H \right) \right) \\
&= \log_2 \left(\det \left(\mathbf{I}_{N_s} + \frac{1}{N_0} \left(\begin{bmatrix} \mathbf{I}_{N_s} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{\Sigma}_{R_H} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{I}_{N_s} \\ \mathbf{0} \end{bmatrix} \sqrt{\mathbf{P}} \right) \left(\begin{bmatrix} \mathbf{I}_{N_s} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{\Sigma}_{R_H} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{I}_{N_s} \\ \mathbf{0} \end{bmatrix} \sqrt{\mathbf{P}} \right)^H \right) \right) \\
&= \log_2 \left(\det \left(\mathbf{I}_{N_s} + \frac{1}{N_0} (\mathbf{\Sigma}_{N_s} \sqrt{\mathbf{P}}) (\mathbf{\Sigma}_{N_s} \sqrt{\mathbf{P}})^H \right) \right) \\
&= \log_2 \left(\det \left(\mathbf{I}_{N_s} + \frac{1}{N_0} \mathbf{\Sigma}_{N_s}^2 \mathbf{P} \right) \right) \\
&= \log_2 \left(\prod_{i=1}^{N_s} \left(1 + \frac{\sigma_i^2}{N_0} p_i \right) \right) \\
&= \sum_{i=1}^{N_s} \log_2 \left(1 + \frac{\sigma_i^2}{N_0} p_i \right)
\end{aligned}$$

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